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MILITARY SPECIFICATION

CONTROL-MONITOR SET, GUIDED WEAPON, AN/AWW-9B, P/N 1064AS4100

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, manufacture, and acceptance requirements for the Control-Monitor Set, Guided Weapon AN/AWW-9B herein referred to as the Pod. The Pod covered by this specification is for airborne use and performs as a major component part of the data link communication network, providing transmission of command signals and reception of video data, between a control aircraft and a WALLEYE guided weapon or Standoff Land Attack Missile. The equipment covered by this specification consists of one major assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Systems Requirements Department, Code SR3, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-W-5088	-	Wiring, Aerospace Vehicle.
MIL-M-7793	-	Meter, Time Totalizing.
MIL-A-8591	-	Airborne Stores, Suspension and Aircraft-Stores Interface (Carriage Phase), General Design Criteria for.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts); Packaging of.
MIL-N-18307	-	Nomenclature and Identification for Aeronautical Systems Including Joint Electronics Type Designated Systems and Associated Support Systems.
MIL-H-46855	-	Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-C-81582/1	-	Connectors, Electric, Bayonet Coupling, Umbilical Receptacle, Jam Nut Mounting, RFI Shielded, Crimp Contact, Class E.
MIL-C-81582/2	-	Connector, Plug, Electric, Bayonet Coupling, Lanyard Release, RFI Shielded, Crimp-Type Contacts, Class E.
MIL-T-81624	-	Television Display Requirements for the Weapon System for Guided Weapon Mark 1 all Mods (WALLEYE) and Guided Weapon, Practice MK 4 Mod 0/1.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-454	-	Standard General Requirements For Electronic Equipment.
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-462	-	Measurement of Electromagnetic Interference Characteristics.
MIL-STD-704	-	Aircraft Electric Power Characteristics.
MIL-STD-781	-	Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-965	-	Parts Control Program.

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STANDARDS (Continued)

MILITARY (Continued)

MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-5400	-	Electronic Equipment, Airborne, General Requirements for.

HANDBOOKS

MILITARY

MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification and Production.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

63A67A30	-	Breathing System.
184AS163	-	Sync Remover and Horizontal Generator Assembly (A4).
184AS166	-	Video Amplifier Assembly (A5).
184AS290	-	Foam Compound.
196AS196	-	SCP Voltage Regulator.
196AS212	-	2.3 kHz Oscillator and Video Driver (A11).
196AS250	-	Power Supply Assembly (A6).
196AS501-3	-	Receiver.
1035AS115	-	BIT No. 1 Assembly (A2).
1035AS120	-	BIT No. 2 Assembly (A12).
1035AS193	-	Mixer, Double Balanced.
1064AS107	-	Casing Assembly.
1064AS141	-	Power Amplifier.
1064AS142	-	Voltage Controlled Oscillator.
1064AS143	-	Low Pass Filter.
1064AS166	-	One Wire Demodulator Assembly.

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DRAWINGS (Continued)

NAVAL AIR SYSTEMS COMMAND (Continued)

1064AS190 - BIT No. 3 Assembly (A13).
1064AS198 - Diplexer.
1064AS204 - Transformer Assembly.
1064AS229 - Multiple Transmitter Assembly.
1064AS232 - Capacitor Assembly.
1064AS233 - Diode Assembly.
1064AS300 - Filter Assembly, Electrical.
1064AS303 - Circuit Breaker.
1064AS327 - Multiplex Filter (A3).
1064AS428 - DPSK Assembly No. 1 (A14).
1064AS433 - DPSK Assembly No. 2 (A15).
1064AS438 - DPSK Assembly No. 3 (A16).
1064AS453 - Logic Decoding Interface (A7).
1064AS462 - Multiplex Decoder and Line Receiver (A10).
1064AS470 - Antenna - Radome Assembly.
1064AS482 - Sync Regenerator Board 1 (A8).
1064AS487 - Sync Regenerator Board 2 (A9).
1064AS500 - Television Receiver Assembly.
1064AS506 - Sample/Hold Video Amplifier Module (B5).
1064AS507 - PJR Circuit/AGC Module (B2).
1064AS508 - Pulse AM Detector Module (B3).
1064AS509 - ±12V Regulator Module (B1).
1064AS608 - Vibration Isolator.
1064AS664 - Transmitter Base/Heat Sink.
1064AS1178 - Power Supply Board Assembly.
1064AS1470 - Antenna - Radome Assembly.
1064AS2139 - Radio Transmitter Assembly.
1064AS2194 - Local Oscillator.
1064AS2281 - End Assembly.
1064AS2292 - Channel Select/BIT Display Assembly.
1064AS2440 - BIT No. 4 Assembly (A17).
1064AS2500 - Television Receiver Assembly.
1064AS2650 - Half Sine Modulator.
1064AS3150 - Transmitter Power Supply Assembly.
1064AS4100 - Guided Weapon Control Monitor Set AN/AWW-9B
1064AS4101 - Interface Control Drawing.
1064AS4109 - Schematic Diagram
1064AS4110 - Signal-Control Processor.
1064AS4391 - SLAM Discrete 5 Control (A1).
1342AS135 - Special Warning Decals.
1342AS180 - Video Cassette Tape Recorder (VCTR).
1342AS510 - De-Emphasis Video Amplifier Module (A15).
1342AS520 - ±12V Regulator Module (A16).

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DRAWINGS (Continued)

NAVAL AIR SYSTEMS COMMAND (Continued)

1342AS531	-	Pre-amplifier (A1); Sample/Hold Module (A12).
1342AS532	-	Filter, Band Pass (A2).
1342AS533	-	Mixer, Microwave (A3).
1342AS535	-	FM Limiter/Discriminator (A14).
184AS290	-	Foam Compound.
196AS196	-	SCP Voltage Regulator.
2300287	-	Cover Assembly, Receptacle.

TECHNICAL MANUALS

NAVAL AIR SYSTEMS COMMAND

NAVAIR 16-30AWW9-3	-	Guided Weapon Control Monitor Set AN/AWW-9B.
NAVAIR 16-30AWW9-3A	-	Guided Weapon Control Monitor Set AN/AWW-9B, Classified Supplement.

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Parts and materials. The parts and materials used in the control-monitor sets shall conform to the requirements of the drawings and subsidiary specifications specified herein, except insofar as errors on the drawings may be incompatible with the requirements of this specification and related specifications. Unless otherwise specified in the contract, proposed changes, either to correct drawing discrepancies or to effect improvements, shall be subject to the acceptance of the contracting activity.

3.2.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts shall be obtained by the contractor in accordance with MIL-STD-965, Procedure I. Approval for use of materials shall be obtained by the contractor in accordance with MIL-STD-5400.

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3.2.2 Supplemental requirements. In the event that the contractor proposes to alter the equipment specified herein and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic conformance. Parts and materials selected for use in the Pod shall conform to the requirements of MIL-STD-5400.

3.3 Design and construction. The Pod shall be manufactured in accordance with the drawings and subsidiary specifications listed in Drawing 1064AS4100 and related lists, except insofar as errors of omission or commission on the drawing may be incompatible with the requirements of this and related specifications. In any case, it is a condition of final acceptance that the Pod manufactured shall meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically accepted by the contracting activity prior to their incorporation.

3.3.1 Physical characteristics.

3.3.1.1 Description. The Pod shall consist of three major sections: The forward radome assembly, aft radome assembly, and the casing (center body) assembly. All principal dimensions and locations of particular features, including center-of-gravity, shall be as detailed in table I and on figure 1.

TABLE I. Weight and dimensions for the Pod.

Parameters	Value
Length	124.3 inches
Diameter 1/	15 inches
Volume	10.3 cubic feet
Weight	690 pounds max

1/ Major diameter of casing excluding bomb lug wells, umbilical connector, heat sink, ejector foot pad and sway brace pads.

3.3.1.2 Total weight. Total weight of the Pod shall be not greater than 690 pounds.

3.3.1.3 Flight requirements. The Control-Monitor Set, Guided Weapon AN/AWW-9B shall be in accordance with Drawing 1064AS4100 and shall be functionally compatible and flight certifiable as an emergency jettisonable Pod on the following aircraft:

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NOTES:
1. STATION (STA) INDICATES DISTANCE IN INCHES FROM NOSE OF FORWARD RADOME.
2. UMBILICAL CONNECTOR PIN ASSIGNMENT GIVEN IN TABLE 1.
3. WEIGHT OF SET IS 690 LBS. (313 KG).
4. VOLUME OF SET IS 10.3 CU. FT. (292 CU. M).

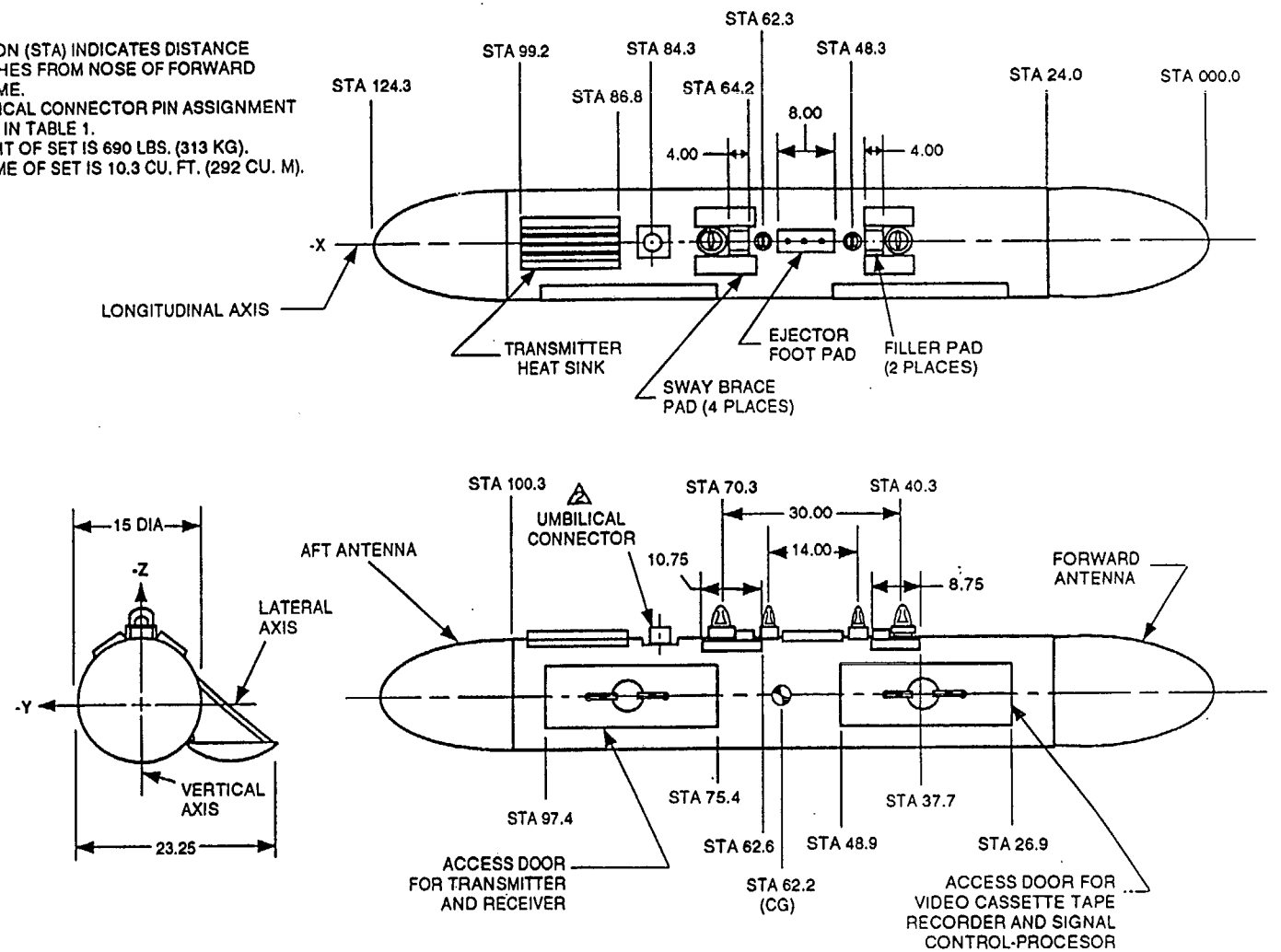


FIGURE 1. Control-Monitor Set, Guided Weapon AN/AWW-9B.

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a. A-4
b. A-7A/B/E
c. TA-7C
d. F/A-18

The AN/AWW-9B Pod shall be designed for carriage on the centerline or inboard wing-station, as applicable to the specified aircraft, and shall be capable of captive flight.

3.3.2 Reliability.

3.3.2.1 Operational stability. The Pod shall operate within the required performance, continuously or intermittently, for a period of not less than 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The Pod shall have a total operating life of not less than 300 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the Pod, and the wearout life of such parts, shall be determined by the contractor.

3.3.2.3 Reliability in mean-time-between-failure (MTBF). The Pod shall have a specified mean (operating) time between failures (MTBF) of 50 hours.

3.3.2.4 Time totalizing meter. The following units shall contain time totalizing meters in accordance with MIL-M-7793:

Unit	Type Meter
a. Transmitter Power Supply 1064AS3150	M7793/6-001
b. Signal-Control Processor 1064AS4110	M7793/6-001

3.3.3 Construction.

3.3.3.1 Cabling and connections.

3.3.3.2 Umbilical connector. Input/output terminations to the Pod for primary power, control signals, and audio/video information shall be made through the umbilical connector, A1W1J1, mounted on top of the casing assembly. The connector shall be in accordance with MIL-C-81582/1 part number type M81582/1E03SA and shall mate with the aircraft umbilical connector MIL-C-81582/2 part number type M81582/02E03PA1. The umbilical connector pin number assignment, function, and current/voltage requirements are specified on Interface Control Drawing 1064AS4101. A receptacle cover assembly in accordance with Drawing 2300287 shall be provided to protect the umbilical connector when the equipment is not in use.

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3.3.4 Controls.

3.3.4.1 Internal adjustments. No adjustments to the Pod shall be required prior to its use with the following exceptions:

a. The 1A/C-2A/C switch on the Transmitter Power Supply must be preset to the desired position.

b. The circuit breaker, CB1 must be placed in the ON position.

c. The FSK/AUTO modulation switch inside the Signal-Control Processor (SCP) must be set to the desired modulation method.

d. The channel select switch on the Transmitter Power Supply must be set to match the channel of the weapon.

e. The Video Cassette Tape Recorder must be loaded with a tape cassette.

3.3.5 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of Control-Monitor Set, Guided Weapon AN/AWW-9B, in accordance with MIL-STD-5400. The drawings listed in Drawing 1064AS4100 shall be used as the standard to measure interchangeability for the AN/AWW-9B Pod.

3.3.6 Interference control. The generation of electromagnetic interference by the Pod and the vulnerability of the Pod to electromagnetic interference shall be controlled within the limits of MIL-STD-461 for Class IA and Class IB equipment. EMI testing shall be accomplished under standard conditions specified in 3.3.12 and in accordance with MIL-STD-462 to demonstrate compliance with the requirements of MIL-STD-461. A complete interference control plan shall not be required.

3.3.7 Maintainability. Maintainability provisions shall be incorporated within the design to provide for service, access and repair of all assemblies. The replacement of major components within the Pod shall be accomplished without soldering and without the need for special tools. Normal routine testing of the Pod shall be accomplished using the special test equipment listed in factory acceptance test procedures, in addition to the standard test equipment normally available in intermediate maintenance shops. Further special test equipment shall be prescribed by the contracting activity as the need develops (see 6.2).

3.3.7.1 Test connections. Normal testing of the Pod shall be accomplished without disassembly with the exception that the FWD and AFT antenna connections shall be disconnected so that RF test equipment may either monitor or supply signals to the antenna feed cables.

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3.3.7.2 Built-in-test. The major components that comprise the Pod shall have built-in-test (BIT) features to check critical parameters.

3.3.7.3 Periodic maintenance. The Pod shall require no periodic maintenance with the exception of the following:

a. The desiccant breather system and desiccant storage bags shall require periodic servicing.

b. The video cassette tape recorder shall require periodic cleaning of the recording heads and tape path to ensure satisfactory recordings.

c. SCP alignment on a 12-month cycle.

d. Modulation Alignment on a 12-month cycle.

3.3.8 Transportability. The Control-Monitor Set, Guided Weapon AN/AWW-9B shall be transportable by air, rail, truck and ship when mounted in a shipping container as specified in 5.1.

3.3.9 Storage life.

3.3.9.1 Packaged condition. The Pod shall be capable of being stored in unsheltered areas when packaged to the requirements of 5.1. The normal storage environmental extremes shall be any dew point between -65 and +71°C.

3.3.9.2 Unpackaged condition. The Pod shall be capable of storage in the unpackaged, assembled condition for not less than two years, in the weapon ready storage area of an aircraft carrier or shore installation, without impairment to form, fit or function. Normal storage environment extremes shall be any dew point from -54 to +60°C. Covers and plugs shall be in place during storage. The desiccant breather system and desiccant storage bags shall be serviced, replaced, or tested periodically as specified in the operation and maintenance instructions manual for the Pod. No realignment of the Pod, or its component parts shall be necessary after storage except as noted in 3.3.7.3.c and 3.3.7.3.d.

3.3.10 Nameplates and identification marking. Serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Identification marking shall be in accordance with MIL-STD-130. The nameplate shall list the following information:

a. Unit name (Control-Monitor Set, Guided Weapon).

b. Unit nomenclature AN/AWW-9B.

c. Contract number.

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d. Manufacturer's code number.

e. Equipment Drawing Number 30003-1064AS4100.

f. Technical Manual 16-30AWW9-3.

g. Contracting activity (Department of the Navy - Naval Air Systems Command).

h. Serial number shall be stamped on the Pod casing with the number specified in the contract.

3.3.10.1 Decals, stencils, nameplates and markings. All decals, stencils, nameplates, and markings shall be affixed to the Pod in accordance with final assembly Drawing 1064AS4100 and all referenced marking specifications, documents, and drawings thereon.

3.3.10.2 Serial numbers. Serial numbers will be assigned by the contracting activity for the Pod under contract or purchase order. The contractor shall serialize sequentially all Pods which are delivered under the contract.

3.3.10.3 Special warning decals. The forward and aft radome assemblies shall be marked on each side surface with warning stencils labeled "NO STEP RADOME". Each radome assembly on both sides shall be marked with warning stencils labeled "WARNING - RADIO FREQUENCY RADIATION HAZARD - a radiation hazard to personnel exists within 48 inches of Pod antenna radomes during Pod command transmission". The stencil shall meet all requirements of Drawing 1342AS135.

3.3.11 Safety. The contractor shall assure that system safety requirements are in accordance with MIL-STD-5400.

3.3.11.1 Flight safety. Flight safety procedures applicable to the associated aircraft and the WALLEYE guided weapon shall be applicable to the AN/AWW-9B Pod.

3.3.11.2 Ground safety. Ground safety procedures that apply to the associated aircraft and the WALLEYE guided weapon shall apply to the AN/AWW-9B Pod except the designated danger areas of radiation. When the Pod is in the high power mode of operation, the high power area is that area 48 inches directly in front of the Pod radomes and outside the entire perimeter of the radomes up to 6 inches. The power density within the danger area can be greater than 10 mw/cm^2.

3.3.11.3 Human performance/human engineering. The AN/AWW-9B and all component parts thereof, shall be designed in accordance with the requirements

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of MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

3.3.12 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

a.	Temperature	Room ambient (25 ±5°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative
e.	Input power voltage	115 ± 1.0 V rms, 400 ± Hz, 3-phase and 28.0 ± 2.0 Vdc

3.3.13 Service conditions.

3.3.13.1 Environmental service conditions (operating). The control-monitor set shall be constructed so that no moving or movable part or assembly can become undesirably free or sluggish in operation, no movable part or control can be shifted in setting, position, or adjustment, and no degradation be caused in the performance below that specified herein when subjected to the environmental service conditions specified in 3.3.13.2 through 3.3.13.11.

3.3.13.2 Vibration. The Pod shall meet the performance requirements of this specification during and after being subjected to the vibration environmental test detailed in 4.5.1. The performance characteristics specified in 3.4 shall apply.

3.3.13.3 Temperature-altitude. The Pod shall meet the temperature-altitude combination of MIL-STD-5400 for Class 1 equipment except the maximum altitude shall be 55,000 feet (see 4.5.2) assemblies, except for the transmitter, are to be tested 15°C higher than the fully assembled Pod (-40 to +70°C).

3.3.13.4 Humidity. The Pod shall operate as required after being subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I. The equipment shall not be damaged as a result of the humidity test specified herein (see 4.5.3).

3.3.13.5 Shock. The Pod shall meet the performance requirements of this specification after being subjected to a shock test in accordance with MIL-STD-810, Method 516, Procedure I, except that the following deviations shall apply (see 4.5.4):

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a. One shock in each direction shall be applied along the vertical axis of the Pod (total of 2 shocks).

b. The acceleration pulse shall be 20G (sin _t/.060) ± 1G.

c. Tolerance limits of the half sine shock pulse configuration shall be in accordance with MIL-STD-810, Method 516, figure 516.4-2 where the peak value (a) is 20G and the nominal duration (D) is 60ms.

3.3.13.6 Sand and dust. Sand and dust requirements shall be as specified in MIL-STD-5400 (see 4.5.5).

3.3.13.7 Fungus. Fungus requirements shall be as specified in MIL-STD-5400 (see 4.5.6).

3.3.13.8 Salt atmosphere. Salt atmosphere requirements shall be as specified in MIL-STD-5400 (see 4.5.7).

3.3.13.9 Explosive conditions. Explosive conditions requirements shall be as specified in MIL-STD-5400 (see 4.5.8).

3.3.13.10 Rain. The Pod shall operate as required after exposure to rainfall conditions equivalent to MIL-STD-810, Method 506, Procedure II. Covers and plugs shall be in place during exposure (see 4.5.9).

3.3.13.11 Acceleration. The Pod shall operate as required while being exposed to the design requirements of MIL-A-8591 for both wing and fuselage mounting. The Pod shall withstand the aerodynamic and inertial captive flight loads specified in MIL-A-8591 for flight regimes up to MACH 1.2 at sea level (see 4.5.10).

3.3.14 Environmental control. Environmental control shall consist of natural circulation of air within the Pod enclosure for dissipating heat generated by the electronic components. Heat shall also be dissipated by conduction through the component mounting bases and the casing to outside ambient air. Humidity in the Pod enclosure shall be controlled by a desiccant breather system. The breather system pressure differential limits shall be not greater than 3 psi.

3.3.15 Service conditions (nonoperating). Service conditions for storage and transportation shall meet the storage life and storage and transit requirements of 3.3.8 and 3.3.9.

3.3.15.1 Warm-up time. The time required for the Pod to warm-up prior to operations shall be not greater than 5 minutes either under standard conditions (3.3.12) or at extreme service conditions (see 3.3.13).

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3.3.16 Input electrical power.

3.3.16.1 Operating power. The Pod shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The power required shall not exceed the specified amounts.

a. AC power (three phase), 115V, 400 Hz, 1,725 watts maximum.

b. DC power, 28V, 2.6 watts maximum.

3.3.16.2 Protection. The Pod shall be capable of withstanding input power variations as imposed by abnormal electric system operations as specified in MIL-STD-704.

3.3.16.3 Degraded performance. Degraded performance will be permitted for Pods operating during the emergency mode of steady state ac power as specified in MIL-STD-704. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electrical system operation.

3.4 Performance characteristics. The performance of the Pod shall be as specified in 3.4.1 through 3.4.3.3 plus any and all performance requirements stated on the drawings listed in Drawing 1064AS4100. Unless otherwise specified herein, these requirements shall apply under both standard (see 3.3.12) and service conditions (see 3.3.13). When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.4.1 Operation. The AN/AWW-9B Pod shall operate from the aircraft power system. The Pod shall receive data link multiplex signals from the aircraft cockpit controls (see table II of ACFT/Pod Interface), transmit control signals to the weapons, and provide to the cockpit monitor the received weapon video, including signals from the weapon acknowledging launch and target tracking, superimposing them onto the received video signal.

3.4.2 Carrier frequencies. The transmitter and receiver carrier frequencies shall be selected between five different frequencies as specified in the contract (see 6.2). The operator shall have the capability of pre-setting these frequencies at the transmitter power supply or remotely, from the aircraft cockpit.

3.4.3 Functional organization. During operation of the Pod, the operator shall have the capability of selecting, from the cockpit, either the forward or aft data link Pod antenna. A diplexer 1064AS198 shall be used to provide isolation between the command transmitter output and the video receiver input to prevent receiver desensitization, allowing use of one antenna for simultaneous transmission and reception of signals.

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3.4.3.1 Video data link signals. Video data link signals, received at the selected antenna, shall be filtered by the diplexer and applied to the video receiver. The receiver shall demodulate the received video signal and relay it to the SCP for processing.

3.4.3.2 Data link Pod command transmitter (Tones Mode). The data link Pod command transmitter shall accept analog weapon commands from the cockpit controller when a TONE channel is selected. The transmitter shall modulate these analog signals onto the command RF carrier and transmit them to the weapon.

3.4.3.3 Data link Pod Transmitter (DPSK MODE). When the cockpit controller is in the DPSK position, discrete and analog tone signals shall be sent to, and accepted by, the Pod command encoder. These signals shall be encoded and sent to the Pod transmitter, modulate the command RF carrier, and be transmitted to the weapon.

3.5 Schematic diagram. The electrical interconnection of components comprising the equipment shall conform to drawing 1064AS4109.

3.6 Interface description.

3.6.1 Electrical interface. All communication between the data link Pod and the control aircraft cockpit shall be accomplished using an umbilical cable connecting the Pod to the aircraft. The AN/AWW-9B umbilical connector (M81582/1E03SA) pin number assignment, function and current/voltage requirements shall be as specified in table II. Multiple frequency tones originating in the cockpit controller shall be used as the Pod control signals, and shall be applied to the Pod through the aircraft wiring. These various tones shall order the Pod to initiate internal Pod functions and transmit specific command signals to the weapon. Table III includes the specific functions, tone frequencies, and driving currents for the multiplex control signals. One wire channel selection is accomplished by frequency modulating the 1.3 kHz tone with the digital code given in table IV. The Pod channels shall be selectable remotely from the aircraft cockpit using one wire channel select or three wire channel select. Remote selection of channels shall be done automatically using the following priorities. If 120 kilohms to ground resistance is present at umbilical pin 13, the Pod will operate using the three wire channel selection (table V). If 120 kilohms to ground resistance is not present at umbilical pin 13, and the 1.3 KHz signal is modulated as noted in table VI, the channel selection shall be made using the one wire channel select signals in accordance with table IV. If the one wire and three wire signals are not present, the Pod will select the channel as denoted on the channel select switch mounted on the transmitter power supply in the Pod. The Pod shall operate from the aircraft power system.

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TABLE II. Aircraft/Pod interface.

Connector Pin No.	Function	Current/Voltage
1	Ground	
2		
3	Multiplex Control (Signal, Current Inputs) Ground	60 Milliamperes (mA) Peak to Peak 25 mA RMS Maximum
4		
5		
6		
7	Trigger discrete 5 (SLAM)	
8	Pod Status Indicator (PSI) Input	22.8 mA Maximum
9		+28 Vdc, 50 mA Maximum
10	Station Select	
11		+28 Vdc, 100 mA Maximum
12	Channel Select Bit 0	
13	Reference Table IV	+28 Vdc, 114 mA Maximum
	Ground (WPN I.D. GND) Station	
14	Audio Input (To VTR)	Required for A/C to Operate
15		10 Vac p-p Maximum Signal into 620 Ohms
	Channel Select Bit 1	+28 Vdc, 92 mA Maximum
16	Reference Table IV	
17	Cockpit Video Coax Shld	
18	Cockpit Video Coax Shld	
19	Ground	
20		
21		
22	Ground	
23	Channel Select Bit 2	+28 Vdc, 92 mA Maximum
24	Reference Table IV	
25		
26	115V, 400 Hz, ϕ A Input	5 Amperes Maximum
27	115V, 400 Hz, ϕ B Input	5 Amperes Maximum
28	115V, 400 Hz, ϕ C Input	5 Amperes Maximum
29	3 ϕ Neutral Input	5 Amperes Maximum
30		
31	Power Change Over (PCO) (VCTR)	A Ground in AN/AWW-9B
32		Electronics, 1 Ampere Maximum
33		
34		
35	Cockpit Video Coax Shld	
36	Cockpit Video	3 Volts p-p Composite Video in
37		accordance with MIL-T-81624 1/ into 91 ohms
	Cockpit Video Coax Shld	
38		
39		
40		
41		

1/ Except horizontal sync pulse width may be 5.2 to 9 microseconds.

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TABLE III. Aircraft/Pod functions and tone frequencies.

Tone (kHz)	Currents (mA)	Functions
1.3 ± 1.42%	.10 ± .015 mA	Antenna Select/one wire channel select
1.7 ± 0.94 %	.10 ± .015 mA	Test
2.3 ± 3.0 %		Pod Status Indicator
5.4 ± .81%	.12 ± .02 mA	Video ON
7.35 ± .73%	.175 ± .02 mA	Video OFF
10.5 ± .75%	.27 ± .025 mA	Command Override
22 ± 8.18%	.70 ± .04 mA	Azimuth
30 ± 8.15%	1.17 ± .06 mA	Elevation

TABLE IV. One wire channel select code.

Channel Select Switch Letter	BIT				RF Channel	Modulation
	MSB	3	2	LSB		
A	0	0	0	1	A	Tone
C	0	0	1	0	C	Tone
E	0	0	1	1	E	Tone
F	1	1	0	0	A	DPSK
H	0	1	0	1	C	DPSK
J	0	1	1	0	E	DPSK
K	1	1	1	1	-	-
L	0	1	1	1	L	DPSK
N	1	1	0	0	N	DPSK

TABLE V. Three wire channel select code.

Umbilical Pin	Channel							
	A	F	C	H	E	J	L	N
13	1	0	0	1	1	0	1	0
16	0	0	1	0	1	1	1	0
24	0	1	0	1	0	1	1	0

1 = +28 v lts dc
0 = Open circuit

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TABLE VI. One wire channel select word format.

Word Bit	1	2	3	4	5	6	7-16
Data	S S	S S	AFT S S ANT	S LSB	S BIT 2	S MSB	SPARE S

Notes:

S - indicates synchronization signal
S = 1300 Hz
1 = 1371 Hz
0 = 1228 Hz

Data Rate = 50 word bits per second

Pod must accept: 1268 < sync < 1332 Hz
1175 < 0 < 1236 Hz
1365 < 1 < 1427 Hz

3.6.2 Mechanical interface. The mechanical interface between the Pod/aircraft shall ensure compatibility with all WALLEYE configured bomb racks, including AERO 7A, AERO 20A, MAU-9A/A, MAU-12B/A BRU-10 and BRU-32. The umbilical connector, bomb lug wells and suspension lugs, and ejector foot and sway brace pads, shall be dimensioned and located as specified on Interface Control drawing 1064AS4101. Both 14- and 30-inch spacing shall be available on the Pod. The ejector footpad shall engage the stores-away switch in the aircraft bomb rack when the Pod is loaded onto the rack.

3.6.3 Pod AN/AWW 9B/test equipment interface. Test equipment interface shall be as shown on the AN/AWW-9B Pod/test equipment interface diagram, figure 2. The input power for flight line or shop test shall be applied to the Pod J1 umbilical connector via the Distribution Box J3425D/AWW, which shall also control the input signals from the AN/DSM-92D Test Set and the tones from the audio generator. The internal components of the Pod shall be easily accessible for test by opening the two side doors. The Pod J1 umbilical connector inputs during test shall be as specified in table II.

3.7 General arrangement. The AN/AWW-9B Pod shall be designed for suspension from a standard WALLEYE-configured bomb rack, and for emergency jettisonable carriage on either the centerline or inboard wing stations for a variety of high performance aircraft.

3.8 Detailed requirements.

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3.8.1 Functional elements. The AN/AWW-9B Pod functional elements shall consist of the following:

a. One Wire Demodulator Assembly PN 1064AS166.

b. Radio Transmitter Assembly PN 1064AS2139.

c. Transmitter Power Supply Assembly PN 1064AS3150.

d. Signal-Control Processor PN 1064AS4110.

e. Television Receiver Assembly PN 1064AS2500. (Alternate: 1064AS500-2)

f. Video Cassette Tape Recorder (VCTR) PN 1342AS180. (Alternate: V-1000AB-R)

g. Antenna-Radome Assembly (2 ea.) PN 1064AS1470. (Alternate: 1064AS470)

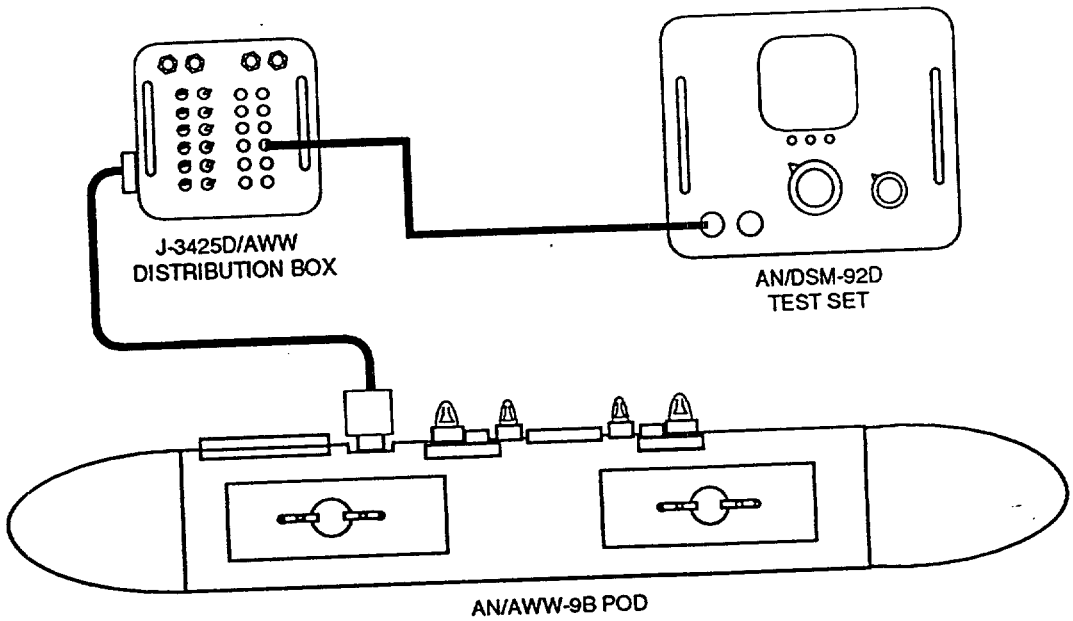


FIGURE 2. AN/AWW-9B pod/test equipment interface diagram.

3.8.2 Radio Transmitter T-1379A/AWW-9. The transmitter for the data link Pod shall transmit coded messages from the control aircraft to the weapon after launch, the weapon to a predetermined target, or if necessary to a secondary target.

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3.8.2.1 Transmitter performance. The data link Pod transmitter shall exhibit the following performance requirements:

a. Transmitter command frequencies shall be as specified in classified supplement NAVAIR 16-30AWW9-3A. The transmitter shall operate on any one of five frequencies selectable from the cockpit or the channel select switch located on the transmitter power supply.

b. Power output of the transmitter shall be not less than 43 dBm (20 watts) measured at the FWD antenna connection, and not less than 44 dBm (25 watts) measured at the AFT antenna connection. Both FWD and AFT power outputs shall have not greater than a 47dB (50 watt) limitation.

c. Transmitter modulation shall be selected within the Pod automatically. Modulation is determined by the channel selected in accordance with table IV.

d. The transmitter shall have a low power mode (120mW) used for Built-in-Test (BIT) testing and for a one-aircraft marriage check with the data link weapon.

e. During the RF Loop portion of BIT, the video signal output shall conform to the composite video parameters of MIL-T-81624 except horizontal sync pulses may be 5.2 - 9 microseconds wide.

3.8.2.2 Transmitter weight. The weight of the transmitter PN 1064AS2139 shall be not greater than 30 pounds.

3.8.2.3 Transmitter content. The transmitter shall contain the following assemblies, subassemblies, circuits, plug-in units, and modules:

a. 1064AS141 Power Amplifier.

b. 1064AS142-2 Voltage-Controlled Oscillator (VCO).

c. 1064AS143 Low Pass Filter.

d. 1064AS2650 Half-Sine Modulator.

e. 1064AS664 Transmitter Base/Heat Sink.

3.8.3 Transmitter power supply PP-7452/AWW-9B. The transmitter power supply shall be capable of supplying 13 amperes at 28 volts dc continuously to the transmitter. Special cooling or heat sinks shall be permitted. The power supply shall provide $\pm$ 12 volts dc to the half-sine modulator and shall provide current limiting and overvoltage protection on the +28 volts dc

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output. The power supply shall contain BIT circuitry to test voltage and current tolerances, VCO and PA BIT Displays, and LO BIT Display.

3.8.3.1 Transmitter power supply weight. The weight of the transmitter power supply shall not be greater than 15.8 pounds.

3.8.3.2 Transmitter power supply content. The transmitter power supply shall contain the following assemblies, subassemblies, circuits, plug-in units and modules:

a. 1064AS1178 Power Supply Board Assembly.

b. 1064AS2292 Channel Select/BIT Display Assembly.

c. 1064AS198 Diplexer, UHF High Power.

d. 1035AS193 Mixer, Double Balanced.

e. 1064AS2194 Local Oscillator.

f. 1064AS2281 End Assembly.

g. 1064AS1178 Power Supply Board Assembly.

h. 1064AS229 Multiple Transmitter Assembly.

i. 1064AS204 Transformer Assembly.

j. 1064AS232 Capacitor Assembly.

k. 1064AS233 Diode Assembly.

l. M3928/15-01 Antenna Select Relay.

3.8.4 Signal-Control Processor C-12333/AWW-9B. The Signal-Control Processor (SCP) shall process the cockpit control signals and supply the signals to the command transmitter modulation circuit for transmission to the weapon. The SCP shall also process the video output from the television receiver and supply video signals to the aircraft cockpit monitor and to the video cassette tape recorder.

3.8.4.1 Signal-control processor performance. Performance requirements for the Pod signal control processor (SCP) shall be as follows:

a. The SCP shall control local application of aircraft power tot he video receiver and video cassette tape recorder.

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b. The SCP shall control the remote selection of the multiplex line.

c. The SCP shall provide regenerated sync and blanking/horizontal drive for a sync comparison test, and provide voltage comparison results for a Phase Locked Loop (PLL) sync regenerator test.

d. The SCP shall provide tone decoding circuits and provide video modulation for the transmitter (only during built-in-tests).

e. The SCP shall remove sync pulses from the television receiver composite video signal, and provide vertical sync and horizontal sync pulses for the sync regenerator control when the weapon video is received.

f. The SCP shall provide automatic gain-controlled video to the video summing amplifier.

g. The SCP shall preamplify the video signal from the receiver and provide automatic gain-controlled video for the preamplifier. The SCP shall also sum the regenerated sync with the video, and provide a composite video output signal to be displayed on the aircraft cockpit video monitor and video cassette tape recorder.

h. The SCP shall supply +28 volts and +12 volts dc for operation of the video cassette tape recorder.

i. The SCP shall provide logic gating for operation of the Pod transmitter, recorder, and antenna switch.

j. The SCP shall provide phase locked loop circuitry to decode tones to control various functions of the Pod as listed in table VII.

k. The SCP shall receive multiplex current signals, convert them to voltage signals and add the 2.3 kHz tone for frequency modulating the transmitter.

l. The SCP shall provide an oscillator circuit to generate the 2.3 kHz tone and provide a video buffer for the video cassette tape recorder.

m. The SCP shall contain built-in-test (BIT) circuitry including: logic circuits for receiving BIT start-stop signals and generating GO and NO-GO signals for front panel light emitting diodes (LEDs); BIT data selector and counter for generating test results and for controlling front panel test fault indicators; BIT decoder, relays, and logic gate circuits for controlling signals when various parameters are to be monitored and rerouted during BIT.

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n. The SCP shall generate the digital code streams for DPSK modulation of the transmitter from the logic signals provided by the multiplex tone decoders.

o. The SCP shall supply controlling signals to the Pod command transmitter to allow a high-power transmission of command signals. During BIT, or when the 1.7 kHz TEST mode is selected the SCP shall allow only low power (≤200mw) transmissions.

3.8.4.2 Signal-control processor weight. The weight of the SCP shall not greater than 14.6 pounds.

3.8.4.3 Signal-control processor content. The signal-control processor shall contain the following assemblies, subassemblies, circuits, plug-in units and modules:

a. 1064AS4391 SLAM Discrete 5 Control (A1).

b. 1035AS115 BIT No. 1 Assembly (A2).

c. 1064AS327 Multiplex Filter (A3).

d. 184AS163 Sync Remover and Horizontal Generator Assembly (A4).

e. 184AS166 Video Amplifier Assembly (A5).

f. 196AS250 Power Supply Assembly (A6).

g. 1064AS453 Logic Decoding Interface (A7).

h. 1064AS482 Sync Regenerator Board 1 (A8).

i. 1064AS487 Sync Regenerator Board 2 (A9).

j. 1064AS462 Multiplex Decoder and Line Receiver (A10).

k. 196AS212 2.3 kHz Oscillator and Video Driver (A11).

l. 1035AS120 BIT No. 2 Assembly (A12).

m. 1064AS190 BIT No. 3 Assembly (A13).

n. 1064AS428 DPSK Assembly No. 1 (A14).

o. 1064AS433 DPSK Assembly No. 2 (A15).

p. 1064AS438 DPSK Assembly No. 3 (A16).

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q. 1064AS2440 BIT No. 4 Assembly (A17).

3.8.4.3.1 Line receiver. The multiplex line shall supply a nominal 6.8 milliampere peak-to-peak current signal consisting of the multiplexed, pre-emphasized, subcarrier oscillator frequencies generated by the cockpit controller. The line receiver shall be a low input impedance amplifier which converts the received signal current to a proportional voltage output. This output shall be summed with the 2.3 kHz oscillator output (see 3.8.4.3.3). The resultant signal shall be applied to the transmitter modulation driver circuit. The driver circuit shall buffer the input signals, set the output voltage level, and apply these signals to the transmitted frequency modulation input terminal which presents a high impedance load. The modulation signal level shall be adjusted at the driver to give a transmitter deviation of 220 kHz peak-to-peak with the 1.7 kHz, 7.35 kHz, 10.5 kHz, 22 kHz, and 30 kHz signals simultaneously applied on the multiplex line.

3.8.4.3.2 Tone detectors and logic decoding. The output signal of the line receiver shall be de-emphasized and detected by phase-locked-loop detectors which have been tuned to each subcarrier oscillator frequency comprising the multiplex signal. The resultant dc voltage from each detector shall be applied to the logic decoding circuitry which shall provide selective gating for operation of the Pod transmitter, video cassette tape recorder, and antenna select relay. Application of the command signal or cockpit control voltage to the SCP shall result in a Pod response as specified in table VII.

3.8.4.3.3 Transmitter output control. The high power mode of the transmitter shall be indicated by illumination of the PA and VCO LED's on the transmitter power supply. Upon application of the TEST (1.7 kHz) command, (Pod in 1-AC Mode) the Pod shall be enabled in low power (10-200 mw) only. Low power radiation shall be indicated by illumination of the VCO LED only. Transmitter output shall be enabled so long as the appropriate command signal is applied from the cockpit, and for 3 (+1, -0) seconds thereafter. At all other times, the transmitter shall be in the Standby mode (-45 dBm or less).

3.8.4.3.4 Arm logic and 2.3 kHz oscillator. Application of +28 volts dc from the cockpit control to the PSI Control Input to the SCP shall energize 2-pole relay A2A11K1. One set of relay contacts shall apply a control signal to the logic decoding circuit which enables the transmitter power output. A second set of relay contacts shall connect the output of the 2.3 kHz oscillator in the SCP to the summing input of the transmitter modulation driver. The oscillator shall supply a 2300 $\pm$7 Hz sine wave with an amplitude of .66 $\pm$ .05 volts peak-to-peak.

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TABLE VII. Command signal and pod response.

Input Signal	Resultant Operation 1/	
	TONE Channel	DPSK Channel
Station Select (+28 Vdc from cockpit)	Energize SCP, VCTR, and Video Receiver	Same as TONE.
Pod Status Indicator (PSI) Control (ITL) (+28 volts dc from cockpit)	Enables transmitter power output and applies 2.3 kHz tone to transmitter modulation driver circuit.	Enables transmitter power output and send PSI code bit for 3 sec minimum.
Aft Antenna Select	(1.3 kHz) Aft antenna connected to Rf circuitry; forward antenna disconnected, DS2 lamp on SCP illuminated.	Same as TONE.
Test Function	(1.7 kHz) Enable transmitter power output in low power mode if 1A/C mode is selected in Pod (high power in 2A/C mode).	Same as TONE.
Video RF ON	(5.4 kHz) Enable transmitter power output; +12 Vdc from SCP applied to VCTR - VCTR advances tape; signal ground connected to PCO indicator wire.	Same as TONE; send DPSK Vid ON bit for 3 sec minimum.
Video RF OFF	(7.35 kHz) Enable transmitter power output; VCTR to standby mode; ground removed from PCO indicator wire.	Same as TONE; send DPSK Vid OFF bit for 3 sec minimum.
Command Override	(10.5 kHz) Enable transmitter power output.	Enable transmitter power output and send DPSK CO bit; transmitter is ON for 3 sec minimum. Bit on code changes immediately.

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TABLE VII. Command signal and Pod response - Continued.

Input Signal	Resultant Operation 1/	
	TONE Channel	DPSK Channel
Azimuth and Elevation	AZ and EL signals sent to the weapon whenever transmitter is enabled.	AZ and EL signals from A/C are digitized and sent in DPSK code stream whenever transmitter is enabled.

1/ Whenever transmitter is on, all multiplex tones received by the Pod are transmitted to the weapon in tone operation.

3.8.4.3.5 PCO indicator and VCTR remote control. Application of the VIDEO RF ON command from the cockpit shall energize 2-pole latching relay A2A7K1. One set of relay contacts shall apply a signal ground termination to the PCO Indicator wire in the umbilical cable, enabling the cockpit PCO (A-7 aircraft) or VTR (F/A-18 aircraft) indicator. A second set of relay contacts shall connect +12 volts dc from the SCP power supply to the video cassette tape recorder, enabling it to advance tape and record. The relay shall remain in the "ON" position although the VIDEO RF ON command is removed. When the VIDEO RF OFF command is applied, the relay shall be reset to the "OFF" position. At this time the cockpit PCO/VTR indicator shall be turned off and the VCTR shall return to the standby mode.

3.8.4.3.6 Aft antenna indicator. Application of the aft antenna select command from the cockpit controller shall result in the logic decoding circuit activating antenna select switch A5K1 in the Transmitter Power Supply end assembly. Transmission and reception of RF signals shall then occur from the aft antenna-radome assembly rather than the forward antenna-radome assembly. Indicator lamp DS2 on the SCP front panel shall illuminate when the aft antenna is selected.

3.8.4.4 Video processor. The signal-control processor shall process the television receiver video output signal and supply video signals to the aircraft cockpit monitor and the video cassette tape recorder. The SCP shall use the received video information and sync pulses generated internally to reconstruct a composite video and video sync signal which has continuous sync and relatively uniform contrast during the presentation. Use of the video processor shall eliminate the most disruptive aspects of a received video signal which may be intermittent, have a wide disparity of signal strength, contain overriding noise and have overlapping or phase shifted sync due to multipath reception.

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3.8.4.4.1 Preamplifier. The received video signal shall be applied to the video preamplifier which shall provide an automatic-gain-controlled output signal. The preamp input impedance shall be 93 ±9 ohms. The output signal strength shall be 1.0 ±0.2 volts peak-to-peak. Composite sync recovered from the sync remover circuit shall be compared to a preset gain control voltage and applied to a feedback circuit which shall control the preamp gain, thus providing the AGC for the received signal.

3.8.4.4.2 Sync remover. A sync remover circuit shall receive the preamp video output and the strip off the composite (horizontal and vertical) sync at a constant level on the signal, that being the video blanking pedestal. A reference voltage generated by a gate array timed to sample only the horizontal sync pulses shall clamp the video input at this level. By tracking the black picture level, the video signal output shall have the correct contrast although the input signal strength may vary considerably.

3.8.4.4.3 Sync stripper and oscillator. A sync stripper shall separate the sync from the preamp composite video output. The vertical sync pulses shall be separated from the composite and sync provided to the sync regenerator circuit. The horizontal sync pulses shall be used to trigger the syncable oscillator which shall provide 15,750 Hz pulses for the sync regenerator. The oscillator shall be adjusted for a free-running frequency of 15,600 Hz with no sync trigger input.

3.8.4.4.4 Sync regenerator. The synchronization (sync) regenerator shall consist of a composite television sync generator which shall be triggered by synchronization pulses derived from the received video signal. A phase-locked-loop crystal-controlled oscillator shall lock the horizontal sync to the syncable oscillator. The vertical sync shall be reset by the vertical sync output of the sync stripper circuit. During periods of intermittently received video signals or noisy sync, the generator shall continue to provide composite sync to the video amplifier. When useable sync is received, the oscillator shall be in phase with it.

3.8.4.4.5 Video amplifier. The video amplifier shall combine the regenerated composite sync with the gain-controlled sync-less video from the sync remover circuit and provide a composite video output signal to be displayed on the aircraft cockpit monitor. The output impedance shall be 93 ±9 ohms. The composite video signal output shall conform to MIL-T-81624. Video resolution shall be a minimum of 400 vertical lines and a minimum of 7 shades of gray shall be differentiated. The specified video output signal shall be present when the video input signal is attenuated by 10 dB and the input signal-to-noise ratio is a minimum of 3 dB, although a decrease in contrast to 4 shades of gray and a decrease in resolution to 200 vertical lines shall be acceptable. During periods of reception when the picture information is lost due to noise, a raster shall continue to be displayed on the monitor until the picture resumes. (Both horizontal and vertical sync shall be present in the video output during these times.)

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3.8.4.4.6 Video buffer. The video signal output from the video amplifier shall be applied to a buffer amplifier which shall provide the video signal source required by the video tape recorder. The buffer shall consist of a current amplifier which shall drive a 75 ohm line. The peak-to-peak video signal measured across a 75 ohm termination shall be not less than 1 volt and shall be not greater than 1.5 volts.

3.8.4.4.7 Video input/output. BNC coaxial receptacles (jacks) shall be mounted on the SCP front panel and shall provide input/output terminations for the video signals in the SCP. Video signals from the receiver shall be applied to the Video-In Connector J3, which shall be wired to the preamplifier. BIT video generated in the SCP shall be accessible from J4. Video signals from the VCTR shall be applied to connector J5 which shall be wired to the BIT circuitry responsible for testing the video signal. Connector J6 shall provide access to the cockpit video wire in the umbilical cable. The sync-processed video signal output from the video amplifier shall be available from connector J7. Cable A2W1 shall normally be connected between J6 and J7, connecting the video signal output of the SCP to the cockpit monitor.

3.8.4.5 Audio. Audio information from the aircraft cockpit shall be applied via the signal control processor to the video cassette tape recorder to be recorded on an audio track accompanying the video recording. A 620 ohm, 0.25 watt resistor shall be connected between the audio input and signal ground to match the impedance of the aircraft intercom system.

3.8.4.6 Power distribution and overload protection. Local application of aircraft power to the SCP voltage regulators, transmitter power supply and television receiver shall be controlled by the filter-circuit breaker. The three phase power lines shall be protected against overload current by a circuit breaker which is in accordance with the requirements of Drawing 1064AS303 and has the fast time delay range. The total load current on any phase shall be not greater than 5 amperes. The circuit breaker, CB1, shall be located on filter-circuit breaker F-1514/AWW-9.

3.8.4.7 Voltage regulators. The SCP voltage regulators shall supply the power requirements of the SCP circuitry and the video cassette tape recorder. They shall operate from the secondary windings of transformer A2T1, built to the requirements of Drawing 196AS196. The aircraft 115 volts, 400 Hz, 3-phase power source shall supply the exciting power to the transformer primary.

3.8.4.7.1 +28 VDC supply. An unregulated +28 volts dc source, operating from the 3-phase secondary winding of transformer A2T1, shall supply power to the tape recorder, the +12 volts dc regulator in the SCP, and a reference voltage source in the BIT circuitry. The +28 volts dc source shall provide a nominal 3 amps current and up to 4 amps when the recorder is first energized. The ripple voltage on the +28 volts dc bus shall be not greater than ± 2 volts.

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The +28 volts dc source shall be protected from overload currents greater than 5 amps by fuse F1, accessible on the SCP front panel. Indicator lamp DS1 on the SCP front panel shall illuminate when the +28 volts dc source is energized.

3.8.4.7.2 +12 volts dc regulator. The +12 volts dc regulator shall operate from the +28 volts dc source and shall supply 1.5 amps nominal current. (This current was included in the current requirement of the +28 volts dc supply.) The SCP shall be protected by overloads greater than 16 volts maximum on the +12 volt bus line.

3.8.4.7.3 +5 volts dc regulator. The +5 volts dc regulator shall operate from a single phase secondary winding in the transformer and shall supply a nominal 750 mA current. The logic components in the SCP shall be protected by voltage overloads greater than 7 volts maximum by zener diode A2CR2 on the +5 volts bus line.

3.8.4.7.4 Negative voltage. A -12 volts dc regulator, operating from a single phase secondary winding in the transformer, shall supply a nominal 380 mA current. A zener diode regulator, operating from the -12 volts dc bus, shall supply -6.8 volts dc at a nominal current of 230 mA.

3.8.4.7.5 Regulator performance. The ripple voltage measured at the output of any regulator shall be not greater than 100 mV under full load conditions. The load regulation shall be ±1 percent or better.

3.8.4.8 Built-in-test. The signal-control processor shall contain built-in-test (BIT) circuitry, including: logic circuits for receiving BIT execute signals and generating GO and NO-GO signals for front panel LEDs, BIT data selector and counter for generating test results and for controlling front panel test fault indicators, BIT decoder, relays, and logic gate circuits for controlling the sequence of signal parameters monitored during the test; and a video signal generator for modulating the transmitter during BIT. BIT shall operate correctly over the temperature range of -20 to +65°C. The BIT shall be capable of testing the functions listed in table VIII.

3.8.4.8.1 Loop test. Bit step 10, the RF loop test shall be accomplished as follows. BIT video, generated by the SCP shall be used to modulate the transmitter. The transmitted signal shall be mixed with the local oscillator, A3Y1, energized during BIT. The mixed signal shall be further attenuated to -78 dBm and routed via the diplexer to the receiver. The received video signal shall be sent back to the SCP video processor which shall compare the received video to the test video. If the received video horizontal sync is of the same frequency and phase as the test video horizontal sync, the SCP shall provide a GO indication. If sync lock is not obtained, a NO-GO indication shall result. BIT step 13 shall permit the SCP loop test circuitry to be tested alone by applying the BIT video output to the received video input connection and testing the video for sync lock with the generated BIT signal.

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TABLE VIII. BIT sequence.

Test No.	Test Title	Circuit Tested
0	BIT READY (+28 volts dc Test)	SCP +28 volts dc Source
1	Voltage Tests (±12, +5, -6 volts dc)	A6 Board
2	1.7 kHz Tone Decoder Test	A7, A10 Board
3	10.5 kHz Tone Decoder Test	A7, A10 Board
4	1.3 kHz Tone Decoder Test	A7, A10 Board
5	5.4 kHz Tone Decoder Test	A7, A10 Board
6	7.35 kHz Tone Decoder	A7, A10 Board
7	2.3 kHz Oscillator Using Tone Decoder	A7, A10, All Boards
8	VCTR Video Test	VCTR
9	Sync Regenerator Test	A8, A9 Boards
10	Loop Test	VCTR, Pod Transmitter, Television Receiver RF Cabling
11	Aft Antenna Test	Pod Transmitter
12	Transmitter OFF NO TEST	
13	Video Amp Test	A5 Board
14	Spare (Automatic GO)	
15	Test Complete (GO/NO-GO)	

3.8.4.8.2 BIT panel. A BIT panel containing the BIT status and counter indicators, and BIT sequencing switches shall be incorporated in the SCP front panel. The BIT RDY lamp shall indicate that the SCP +28 volts dc source is operating and the Pod is ready for BIT. The AUTO START switch shall be a pushbutton switch and shall initiate automatic sequencing through the 15 test steps. Each test number shall be indicated by the correct combination of illuminated lamps 8, 4, 2, and 1. The GO lamp shall illuminate at the end of each successful test step. The NO-GO lamp shall illuminate and the test sequence shall automatically halt if a test is failed. The failed test step shall be indicated by the illuminated counter Lamps. If the failure is

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corrected, BIT shall re-commence. To proceed after an uncorrected NO-GO indication, the RESTART pushbutton switch shall be pressed. If NO-GO occurred during any step sequence, and the cause of the NO-GO indication is uncorrected, NO-GO shall remain lit after the completion of step 15. It shall be possible to step through the BIT sequence by pressing the MANUAL START pushbutton switch. BIT shall advance one step each time MANUAL START is pressed.

3.8.4.9 DPSK encoder. The DPSK encoder, boards A14, A15, and A16, receives the discrete commands from the tone decoder board, A10, and the two analog signals (azimuth and elevation) and encodes them into a digital code stream. This serial digital signal is then used to shift the phase of the transmitter carrier.

3.8.5 Television Receiver R-2095A/AWW-9. The video receiver for the data link Pod shall receive an RF input signal transmitted from the weapon. The Pod receiver shall meet the following performance requirements:

a. The receiver shall receive an RF input signal consisting of a conventional FM signal plus additive RF jamming pulses.

b. The output of the receiver shall consist of the demodulated FM signal with pulsed interference.

c. The interfering pulse shall be positive going if the received interference signal frequency is above the desired signal frequency, and shall be negative going if the frequency is below the desired frequency.

d. An antijam video processor shall be required as part of the video receiver to minimize the effects of pulsed RF interference on or near the received video frequency. The video processor shall detect, blank, and hold jamming pulses from the receiver video output.

3.8.5.1 Receiver weight. The weight of the video receiver shall be not greater than 23 pounds.

3.8.5.2 Receiver content.

3.8.5.2.1 The video receiver (PN 1064AS500-2) shall contain the following assemblies, subassemblies, circuits, plug-in units and modules.

a. 1064AS506 Sample/Hold Video Amplifier Module (B5).

b. 1064AS507 PJR Circuit/AGC Module (B2).

c. 1064AS508 Pulse AM Detector Module (B3).

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d. 1064AS509 ±12V Regulator Module (B1).

e. 196AS501-3 Receiver.

3.8.5.2.2 The video receiver (PN 1064AS2500) shall contain the following assemblies, subassemblies, circuits, plug-in units and modules.

a. 1342AS531 Pre-amplifier (A1).

b. 1342AS532 Filter, Band Pass (A2).

c. 1342AS533 Mixer, Microwave (A3).

d. 1342AS520 ±12V Regulator Module (A16).

e. 1342AS531 Sample/Hold Module (A12).

f. 1342AS535 FM Limiter/Discriminator (A14).

g. 1342AS510 De-Emphasis Video Amplifier Module (A15).

h. 1064AS507 PJR Circuit/AGC Module (B2).

i. 1064AS508 Pulse AM Detector Module (B3).

3.8.6 Video Cassette Tape Recorder RO-570/AXQ or RO-518/AXQ (alternate). The video cassette tape recorder (VCTR), A4, shall be a rugged, compact recorder that will record the received video presentation during airborne conditions. Simultaneously with the video recording, aircrew comments from the aircraft intercom system shall be able to be recorded on an audio track on the tape. The recorder shall be accessible for tape cassette replacement without requiring downloading of the Pod from the aircraft, or removal of the Recorder from the Pod.

3.8.6.1 Operating environment. The recorder shall operate under all service conditions to which the Pod is subjected. The operating temperature range of the recorder, however, shall be limited to between -20 and +60° Celsius. The recorder shall not be damaged when operated in the standby mode, considered as recorder energized but not advancing tape, over the temperature range of -55 to +71° Celsius.

3.8.6.1.1 Vibration isolation. The recorder shall be mounted on a base which is isolated from the Pod casing assembly to reduce the effects of shock and vibration on the recording equipment. Four friction damped vibration isolators, rated at 6 pounds force, and meeting the requirements of Drawing 1064AS608, shall support the recorder, one at each corner, as it is mounted in the Pod. Independent of the Pod, the recorder with shock mounts attached,

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shall operate as required when subjected to a vibration input of 0.01 g^2/Hz from 10 to 2000 Hz in each of 3 mutually perpendicular planes. Sideways picture motion during vibration shall be acceptable, but there shall be no loss of sync.

3.8.6.2 Physical dimensions. The recorder shall be no greater than 7 inches high, 12 inches wide, and 13 inches deep. Weight shall be not greater than 24 pounds (including covers and connector). The recorder shall have a carriage handle which is not included in the above dimensions.

3.8.6.3 Input/output connector. All input electrical connections to the recorder shall be made through connector J2, mounted on the SCP. Connector J2 shall be located on the SCP. The video output of the recorder shall be connected to the SCP VIA connector J5, mounted on the SCP. The Pod shall be built to allow the use of many different types of recorders. Connector J2 pin function assignment is specified in table IX.

TABLE IX. Pin function assignment.

Pin Number J2	Function
1	Video output
2	Video output shield
3	+28 volts dc
4	Audio output shield
5	Tape unthread
6	
7	Ground
8	Audio output
9	
10	
11	Recorder remote control
12	+12 volts dc power
J5	VCTR video input (for BIT Step 8)

3.8.6.4 Power. The recorder shall operate from +12 for +28 volts dc supplied by the signal-control processor. The power requirement shall be not greater than 80 watts.

3.8.6.5 Controls. Upon application of power, the recorder shall be energized, but shall not advance tape. To advance tape and record a signal, +12 volts dc shall be applied to the RECORD REMOTE control pin in the connector. The recorder shall automatically unthread or shut off when all of the tape has been recorded.

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3.8.6.6 Video recording system. The video recording system shall be a rotary, two-head, helical scan system, using a full field, composite video signal based on American television standards. Recording time shall be not less than 27 minutes.

3.8.6.7 Input video. The input signal shall be 1 volt peak-to-peak ±0.5 volt composite video as measured into 75 ohm load. Sync shall be in the negative-going direction with respect to the video signal. The sync-only component of a signal shall have an amplitude of not less than 0.27 volts and not greater than 0.40 volts.

3.8.6.8 Output video. The output signal shall be 1 volt peak-to-peak, negative-going sync, into a 75 ohm load. Resolution for the output video shall be not less than 7 shades of gray and 200 vertical lines. The signal-to-noise ratio shall be greater than 30 dB. The recorder shall have automatic gain control (AGC) such that a 10 dB decrease in input signal strength shall not adversely affect the video output signal.

3.8.6.9 Audio. The audio signal level input to the recorder shall be 0.5 volt nominal into 30k ohms. The playback audio output signal shall be 0.5 volt for 1 kHz into 1k ohm. The frequency response shall be 100 to 10,000 Hz ±3 dB. The signal-to-noise ratio shall be 40 dB.

3.8.6.10 BIT. If specified in the contract (see 6.2), the recorder shall contain BIT circuitry which shall monitor the following parameters:

a. dc input power applied.

b. Video record mode circuits energized.

c. Recorder servo system sync in phase with the applied video sync.

The latter test shall verify that the recorder video head drum is rotating at the correct speed. LED indicators for each test parameter may be located on the front side of the recorder adjacent to the control panel and shall illuminate to provide a GO indication when the recorder is energized and ready to record input video signals.

3.8.7 Antenna-Radome AS-3504A/AWW-9. The antenna-radome shall be manufactured in accordance with Drawing 1064AS1470 and shall meet the requirements specified in 3.8.7.1 through 3.8.7.3.

3.8.7.1 Functional description. Either the forward antenna, E1, or the aft antenna, E2, may be selected for communication between the data link Pod and weapon. The forward antenna shall provide continuous communication with the weapon after release and within the fore hemisphere of the Pod. The aft antenna shall communicate with the weapon when it is located within the aft

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hemisphere of the Pod. The selected antenna shall receive the video signals transmitted by the weapon and radiate telemetry signals to the weapon. The antenna shall be connected by coaxial cable to the diplexer in the casing assembly via the antenna select relay A5K1.

3.8.7.2 Physical description. The antenna-radome assembly shall consist of a fiberglass radome which contains a mechanically fixed, helical antenna. The helical antenna element shall project forward from a baseplate which forms the antenna ground plane. The antenna base shall be secured to the radome bulkhead. The radome with the antenna mounted inside shall become an enclosed assembly preventing radio interference transmission or leakage into the Pod casing assembly. The interior of the antenna-radome assembly shall be filled with a foam compound in accordance with Drawing 184AS290. The cured foam shall have a uniform density of 2.00 $\pm$ 0.5 lbs/ft^3, and a dielectric constant of not greater than 1.6. Weight of the antenna-radome assembly shall be not greater than 24 pounds.

3.8.7.3 Antenna characteristics. The antenna shall use the nominal gain and beamwidth and maximum side lobes as specified for both the command and video RF frequencies in the classified procuring document. The antenna shall be circularly polarized with the main lobe on the longitudinal axis. The side lobes shall be down a minimum of 10 dB. The VSWR shall be maximum of 1.6:1 and the impedance shall be 50 ohms. The antenna shall contain an impedance matching device to meet this requirement.

3.8.8 Casing assembly. The casing assembly, A1, shall be manufactured in accordance with Drawing 1064AS107 and shall contain solid state electronic components, wiring cables, and associated mounting bases and hardware.

3.8.8.1 Configuration. The body contour of the casing shall be continuous with the surface of the forward and aft antenna-radome assemblies where they are joined together. Internal arrangement and construction of the casing assembly shall permit removal and reinstallation of the component equipment contained therein through the access doors, or access area at heat sink mounting area.

3.8.8.2 Access doors. Access to the components within the Pod shall be accomplished through the two access doors in the casing assembly. The surface contour of the access doors, including fasteners and fastening hardware, shall be located within the major diameter of the casing assembly. The doors shall be sealed against moisture penetration around the opening and shall be electrically grounded to the casing to minimize transmission of electromagnetic radiation through the opening. Each door shall be marked with directions on how to open and close the door.

3.8.9 Filter-circuit breaker F-1514/AWW-9. The filter circuit breaker assembly, A6, shall be manufactured in accordance with Drawing 1064AS300. It

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shall filter the aircraft AC power lines at the input to the equipment to prevent broadband radio frequency interference from being conducted to the equipment within the requirements of MIL-STD-461 from Class I equipment.

3.8.10 Signal Demodulator MD-1114/AWW-9 (one wire demodulator). The one wire demodulator shall demodulate the digital FM 1.3 kHz tone generated in the aircraft control group converter to provide a 4-bit channel select code to the transmitter power supply, an antenna select code to the SCP, and spare discrete codes that are currently unused.

3.8.11 Breather system. A breather system meeting the requirements of Drawing 63A67A30 shall be installed in the lower aft end of the casing assembly.

3.8.12 Desiccant bags. Desiccant MIL-D-3464, type I, eight, eight-unit bags, shall be enclosed in a retainer located at the forward end of the casing assembly to protect the video cassette tape recorder from humidity and moisture condensation.

3.8.13 Wiring. Wiring shall meet the requirements of MIL-W-5088 except hook-up and harness wiring may be unmarked and uncoded.

3.8.14 Workmanship. Workmanship for the Pods shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All Pods shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance with requirements; however, this does not authorize the submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection. First article tests shall be conducted by the contractor on a Pod representative of the production Pods to be supplied under the contract. First article tests shall be accomplished under the approved test procedure of 4.6. No first article inspections shall be conducted prior to approval of the first article test procedure by the contracting officer.

4.3.1 Scope of first article tests. First article tests shall include all tests deemed necessary by the contracting activity to determine that the Pod meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-STD-810 electromagnetic interference tests in accordance with MIL-STD-461.

4.3.2 First article approval. Approval of the first article sample will be by the contracting activity upon satisfactory completion of all tests and first article report being accepted. No production Pods shall be delivered prior to the approval of the first article sample.

4.3.3 Production equipments. Pods supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equal to the approved first article sample. Each Pod shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of noncompliance with the above shall constitute cause for rejection of a Pod and of Pods already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the contracting activity.

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing shall be under the supervision of the Government inspector. The contractor shall retain test results documentation for all quality conformance tests. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.4.1)

b. Sampling tests (see 4.4.2)

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4.4.1 Individual tests. Each Pod shall be subjected to the individual tests. These tests shall be acceptable to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each Pod accepted shall pass the following tests:

a. Examination of product (see 4.4.1.1)

b. Operational test (see 4.4.1.2)

c. Manufacturing run-in test (see 4.4.1.3)

d. Workmanship vibration (see 4.4.1.4)

e. Evaluation of test (see 4.4.1.5)

f. Environmental burn-in-test (see 4.4.1.6)

4.4.1.1 Examination of product. Each Pod shall be examined carefully to determine that the material, interchangeability, workmanship, and safety requirements specified herein have been met.

4.4.1.2 Operational test. Each Pod shall be operated long enough to stabilize and to check sufficient characteristics and record data to assure specific equipment operation at room ambient in accordance with the accepted test procedure.

4.4.1.3 Manufacturing run-in test. Each Pod shall be operated under the conditions of workmanship vibration and environmental burn-in specified herein without failure. A failure shall be considered as anything which causes malfunctioning of the Pod. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during normal use of the Pod. Workmanship vibration shall precede the environmental burn-in test.

4.4.1.4 Workmanship vibration. Each Pod shall be vibrated for a period of 15 minutes in the vertical axis (Z-axis as shown in figure 1) using the shaped spectrum shown on figure 3. The Pod shall be operated during this vibration period for the purposes of detecting flaws and imperfect workmanship. Should a failure occur, it shall be repaired and the test started over. The forward and aft doors may be removed from the Pod in order to obtain access to the components and cables within the Pod during vibration.

4.4.1.5 Evaluation of test. During vibration some snow interference and wavy motion on the video presentation is permissible. Tearing of the picture or black streaks in the video presentation shall be cause for rejection. The video strength shall not be affected during the vibration more than ± 1 dB referenced to the signal source. The video quality shall not be affected when

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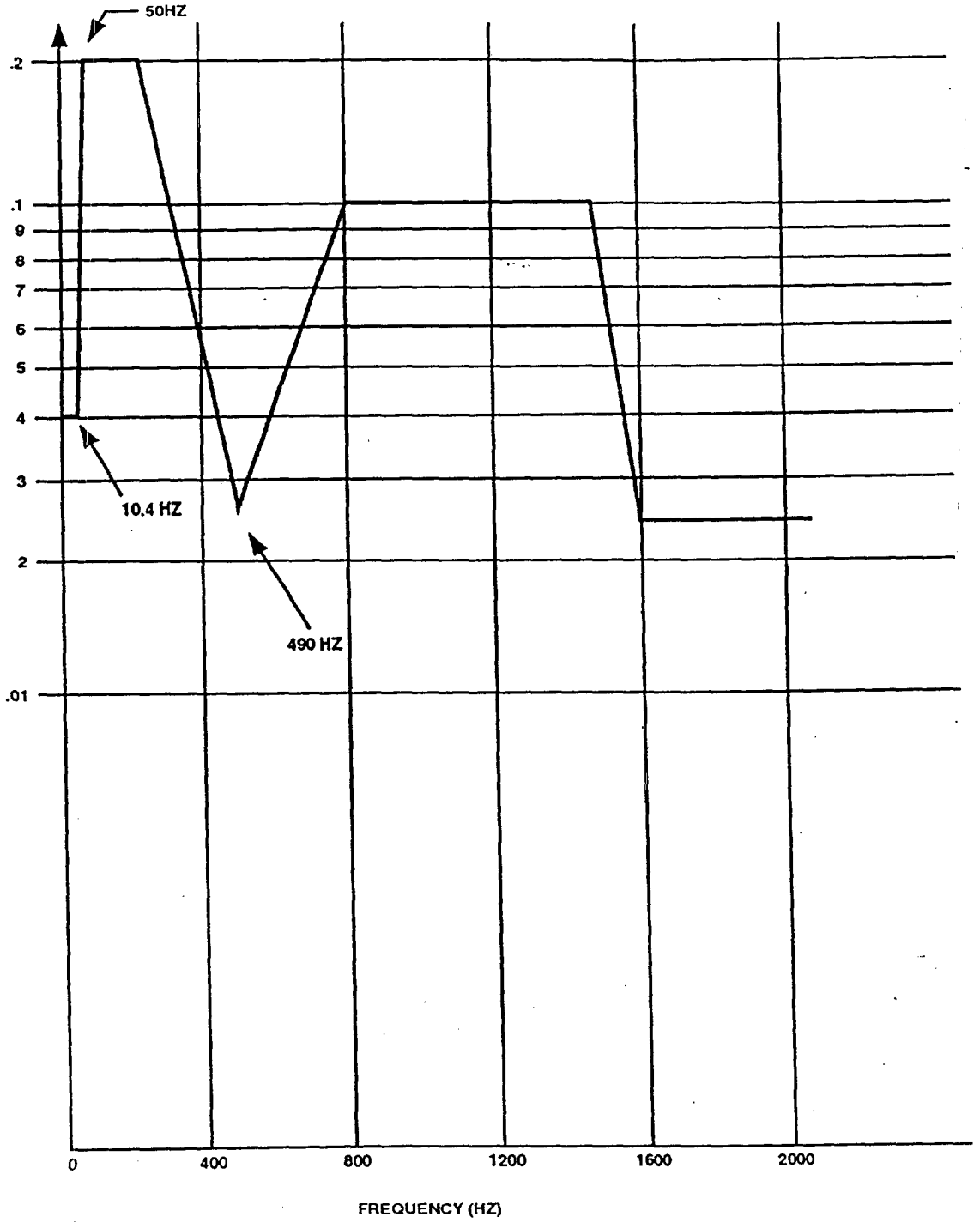


FIGURE 3. Random vibration power spectral density profile.

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command functions are energized. The transmitted command signal shall not be interrupted during the vibration exposure. Proper Built-In-Test (BIT) operation is not required during vibration; however, the BIT operation shall be normal after vibration. The video tape presentation can exhibit noise streaks in the video presentation, waving motion, and intermittent loss of sync during the vibration mode; however, a video tape presentation of video signals recorded after vibration shall appear normal.

4.4.1.6 Environmental burn-in-test. Each Pod shall be subjected to seven 12-hour environmental test cycles as shown on figure 4. Each test cycle shall consist of a 7-hour period of continuous Pod operation preceded by a 5-hour cold temperature stabilization period. The test cycles shall include temperature cycling from -54 to +55°C. The radomes may be removed from the Pod to accommodate the units in test to the test chamber. Each Pod in test shall be subjected to an electrical performance test at +55°C as well as a brief performance check at -54° once daily. At -54° immediately following energization, each Pod in test shall be monitored for proper video presentation on one of the five channels (A,C,E,L, or N). A different channel shall be monitored each day such that all five channels are evaluated periodically during the test. Only Pod operating time is counted as test time. Testing shall continue until each Pod has successfully accumulated 49 hours of environmental test time, the last 14 of which must be failure free. A record of failures occurring during the environmental burn-in shall be maintained as the official record of the progress of the environmental burn-in time. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future Pods. The environmental burn-in test shall not be required for a Pod that is subjected to the reliability assurance test.

4.4.2 Sampling tests. Pods selected for sampling tests shall first have passed the individual tests. Pods will be selected for sampling tests by the Government inspector in accordance with table X.

TABLE X. Sampling tests sample selection.

Quantity of equipment offered for acceptance	Quantity to be selected for sampling test
First 10 Next 50	1 1/ 1 1 for each additional 50 or fraction thereof

1/ Sampling tests shall not be required when reliability assurance tests are conducted.

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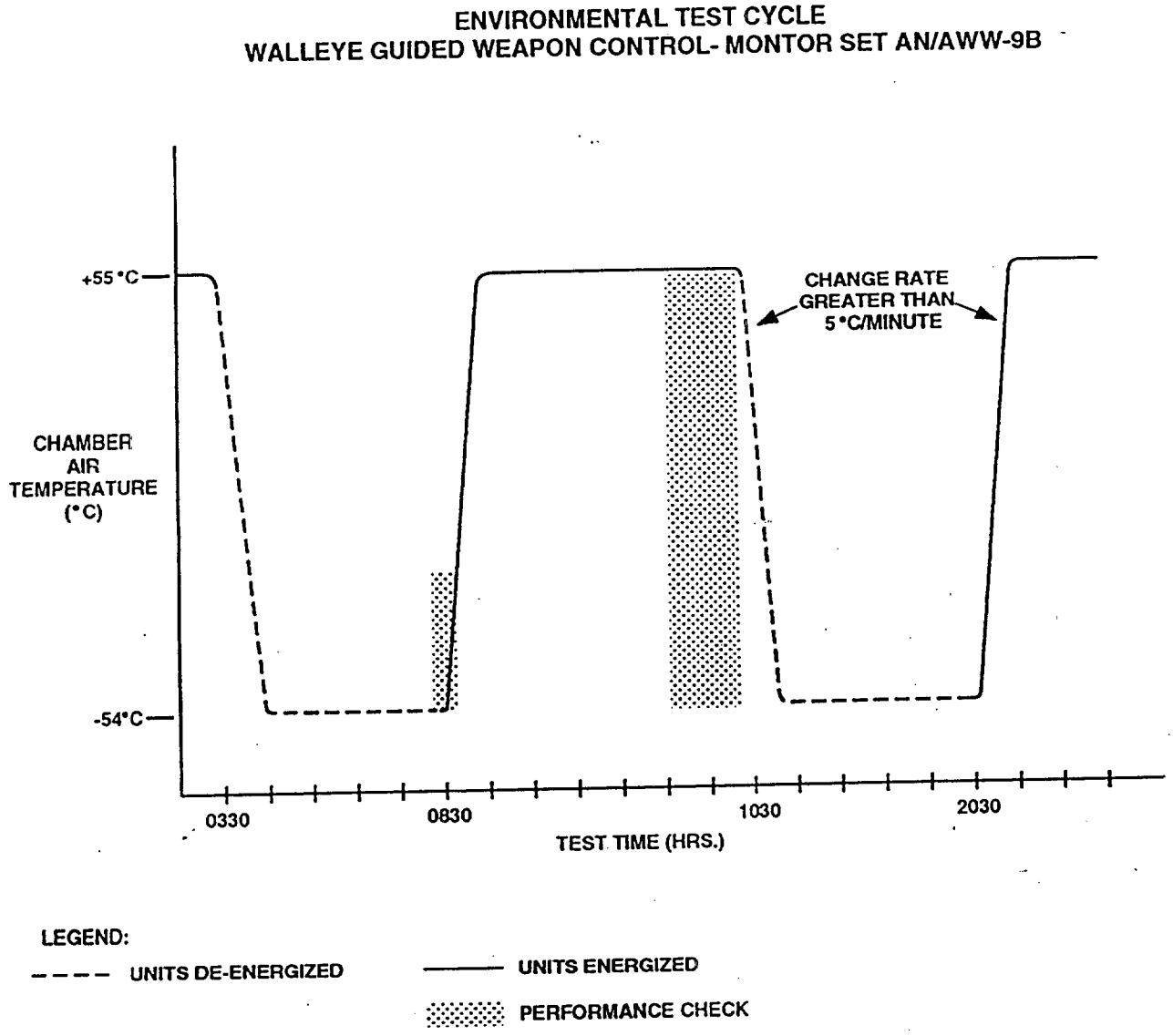


FIGURE 4. Reliability evaluation test.

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4.4.2.1 Scope of tests. As a minimum, each Pod selected for sampling tests shall be subjected to the following tests:

a. Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational tests at certain environmental conditions. The conditions may vary for each Pod tested and should be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.4.1.3 except that the environmental burn-in test duration shall be 300 hours with no restriction on the number of failures. However, each failure shall be analyzed as to the cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.4.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the reliability qualfication phase (see 4.4.3.1) and the reliability production acceptance phase (see 4.4.3.2) shall be conducted. Reliability test plan and reliability test procedures shall be prepared prior to the conduction of the tests and approved by the procuring activity.

4.4.3.1 Reliability qualification phase. Prior to acceptance of equipments under the contract or purchase order, 2 of the first 3 production units shall be subjected to reliability qualification test in accordance with MIL-STD-781, Task 301, consisting of 12-hour environmental test cycles. Testing will continue until a total of 1000 hours of operating time has been accumulated on the 2 units, or approximately 500 hours per Pod. The radome may be removed from the Pod to accommodate the units in the test to the test chamber.

4.4.3.1.1 Test procedure. Each test cycle shall consist of a 5-hour temperature stabilization period at -54°C followed by a 7-hour period of continuous Pod operation at +55°C. Input voltage cycling shall not be required. Each Pod in test shall be subjected to an electrical performance test at +55° as well as a brief performance check at -54°C once daily (as shown in figure 4). At -54°C immediately following energization, each Pod in test shall be monitored for proper video presentation on one of the five channels (A, C, E, L, N). A different channel shall be monitored each day, such that all five channels are evaluated periodically during the test. Once weekly, each Pod in test shall be subjected to a complete operational test at room ambient conditions. At the completion of the 1000 hours test time, each Pod shall undergo a final operational test under standard conditions.

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4.4.3.1.2 Evaluation of tests. The total test time and total relevant failures accumulated during the reliability qualification tests shall be used to determine compliance with the specified Pod MTBF. Only Pod operating time is counted as test time. Specifically, the test MTBF shall be calculated by dividing the total test hours by the number of relevant failures. If the test MTBF is equal to or greater than the specified MTBF for the Pod and no pattern failures are present, the Pod is acceptable. If the test MTBF is lower, or if pattern failures exist, the Pod is rejected and corrective action shall be accomplished. Reliability test reports on failure actions and corrective actions shall be prepared in accordance with MIL-STD-781.

4.4.3.1.3 Reconditioning of reliability qualification test samples. On completion of the reliability qualification test, the Pod shall be reworked by the contractor by replacing all worn or damaged items, including finish and markings. The worn items shall be determined by agreement between the contractor and the contracting activity. After reworking, the contractor shall make available the reworked Pod for quality conformance tests limited to visual examination and the operational test under standard conditions.

4.4.3.2 Reliability production acceptance test. Each Pod produced, other than those subjected to the reliability qualification test, shall be subjected to the reliability acceptance test in accordance with MIL-STD-781, Task 302. The environmental burn-in test, in accordance with 4.4.1.3, shall be the basis from which accept/reject criteria for production reliability is determined.

4.4.3.2.1 Pre-test burn-in. A burn-in (or debugging) period prior to the test period of 4.4.1.3 may be used at the option of the contractor (see 6.2). If an additional burn-in period is to be used, the details thereof must be included in the approved test procedures. If used, the pretest burn-in must be the same for each Pod in time and severity. Failures occurring during this period may be considered nonrelevant but shall be recorded.

4.4.3.2.2 Evaluation of test. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting pretest burn-in operating time and failures) shall be totaled and the results compared with the reject line of Test Plan II of MIL-HDBK-781. These totals shall accumulate so that any time the experience from the beginning of the contract is included. At any one time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the contracting activity shall be notified. The contracting activity reserves the right to stop the acceptance of Pods at any time that a reject situation exist pending a review of the contractor's effort to improve the Pods, the Pod parts, or the Pod workmanship, so that the entire compilation will show other than a reject decision.

4.4.3.2.3 Disposition of tested Pods and report requirements. Each Pod completing the test shall be subjected to a visual examination and final operational test under standard conditions.

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4.4.3.3 Test details. The test details shall be part of the test procedures approved by the contracting activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.4 Special test. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the Pod and to assure adequate quality control. The Pod selected for special tests may be selected from Pods previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule. Selection of Pods for special tests shall be made as follows:

a. On any early Pod after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the contracting activity.)

4.4.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the contracting activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for approval prior to conducting the tests.

4.4.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or a design defect.

c. Where practical, include a test in the individual test to check all Pods for the requirement until assurance is obtained that the defect has been corrected.

4.5 Environment tests. The Pods shall operate as required when subjected to the tests specified in 4.5.1 through 4.5.1.3.

4.5.1 Vibration. The Pods shall meet the performance requirements of this specification during and after being subjected to the vibration environment specified in 4.5.1.1 through 4.5.1.3.

4.5.1.1 Vibration test set-up. The equipment shall be mounted and tested with its longitudinal axis horizontal and positioned so that its mounting lugs are on top of the center body section. The Pod shall be hand mounted directly to the shaker using a cradle to support the center body section. The

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stiffness of the mounting fixture shall be such that its induced resonant frequencies are as high as possible but none shall be less than 800 Hz.

4.5.1.2. Random vibration test. The Pod shall be subjected to 4 hours of random vibration along each of the 3 mutually perpendicular axes of the equipment. The power spectral density of the test control signal shall be 0.006 G^2/Hz over the frequency range from 10 to 1200 Hz and 0.022 G^2/Hz from 1201 to 2000 Hz.

4.5.1.3 Random vibration equalization. The accelerator spectral density applied to the Pod, as determined by an analyzing filter with a bandwidth not greater than 25 Hz between 20 and 200 Hz, shall be ±2 dB over the frequency range from 10 Hz to 2000 Hz; all peaks and notches shall be within the ±2 dB range. Tolerance levels in terms of dB are specified as follows:

$$dB = 10 \log \frac{G_1^2/Hz}{G_0^2/Hz} \text{ or } dB = 20 \log \frac{G_1}{G_0}$$

Where G_1^2 equals the acceleration power spectral density and G_1 equals Grms over the analyzer bandwidth. The term G_0 shall be the specified level. (RMS levels shall be measured using a true RMS meter.)

4.5.2 Temperature-altitude. The Pod shall be subjected to temperature-altitude combination requirements of MIL-STD-5400 for Class 1 equipment except the altitude shall be not greater than 55,000 feet.

4.5.3 Humidity. The Pod shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I.

4.5.4 Shock. The Pod shall be subjected to a shock test in accordance with MIL-STD-810, Method 516, Procedure I, except that the following deviations shall apply. One shock in each direction shall be applied along the vertical axis of the equipment (total of two shocks). The acceleration pulse shall be 20g (sin t/.060) ±1g. Tolerance limits of the half sine shock pulse configuration shall be in accordance with MIL-STD-810, Method 516, figure 516.4-2, where the peak value (A) is 20g and the nominal duration (D) is 60 ms.

4.5.5. Sand and dust. The Pods shall be subjected to sand and dust exposure as specified in MIL-STD-5400.

4.5.6 Fungus. The Pods shall be subjected to fungus exposure as specified in MIL-STD-5400.

4.5.7 Salt atmosphere. The Pods shall be subjected to explosive conditions as specified in MIL-STD-5400.

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4.5.8 Explosive conditions. The Pods shall be subject to explosive conditions as specified in MIL-STD-5400.

4.5.9 Rain. The Pod shall be subjected to rainfall conditions equivalent to MIL-STD-810, Method 506, Procedure II. Covers and plugs shall be in place during exposure.

4.5.10 Acceleration. The Pods shall be exposed to the design requirements of MIL-A-8591 for both wing and fuselage mounting. The Pods shall withstand the aerodynamic and inertial captive flight loads specified in MIL-A-8591 for flight regimes up to MACH 1.2 at sea level.

4.6 Test procedures. The procedures used for conducting first article tests and quality conformance tests shall be prepared by the contractor. The right is reserved by the contracting activity or the Government inspector to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract.

4.6.1 Failure report. In the event of a failure, the Government inspector shall be notified immediately.

4.7 Reconditioning of tested Pods. Pods which have been subjected to quality conformance tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall make available the reworked Pods for acceptance.

4.8 Presubmission testing. No item, part or complete Pod shall be made available by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

4.9 Rejection and retest. Pods which have been rejected may be reworked or have parts replaced to correct the defects and again made available for acceptance. Before making available the reworked Pods, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be prepared.

5. PACKAGING

5.1 General. The Pod shall be packaged in a Mk 426 Mod 0 shipping and storage container in accordance with the MIL-E-17555, Preservation and Packaging Level A, Method IIa. The packaged Pod shall be able to withstand storage, rehandling, and reshipment without necessity of repackaging.

5.2 Marking for shipments. Shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The data link Pod, Control-Monitor Set, Guided Weapon AN/AWW-9B, is a major component part of the WALLEYE data link and Standoff Land Attack Missile weapon systems and provides two-way communication between a control aircraft and a guided weapon. The Pod is designed for suspension from a standard WALLEYE- configured bomb rack, and for emergency jettisonable carriage on either the centerline or inboard stores stations for a variety of high performance aircraft. Weapon control is accomplished from the cockpit via employment of a control unit and video monitor. The Pod receives cockpit control signals and transmits them as encoded RF signals from the Pod antenna to the weapon. Video signals from the weapon are received by the Pod and processed for display on the cockpit video monitor. The data link Pod will permit the control operator to monitor the flight of the weapon toward a target area and, if necessary, command the weapon to change its lock-on point after launch.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Selection of applicable levels of preservation and packing (see Section 5).

d. When a first article is required for inspection and approval (see 4.3).

e. When further special test equipment is prescribed (see 3.3.7).

f. Carrier frequency selection (see 3.4.2).

g. When pretest burn-in is used (see 4.4.1.3).

h. When recorder BIT is prescribed (see 3.8.6.10).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 227.405-70 exempts the requirement for a DD Form 1423.

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Paragraph No.	DID No.	DID Title	Option
4.4.3, 4.6	DI-RELI-80251	Procedure, Reliability Test	----
3.3	DI-CMAN-80639A	Engineering Change Proposal (Long Form)	----
4.4.1.5, 4.6.1	DI-R-21598	Report, Failure	----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the Pod should be a first article sample. The first article should consist of one Pod. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement of samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Abbreviations. Abbreviations used in this specification are listed below:

a. AGC Automatic gain control
b. BIT Built-In-Test
c. DPSK Differential phase shift keyed
d. FSK Frequency shift keyed
e. ITL Intent-to-launch
f. LED Light emitting diode
g. LO Local oscillator
h. PA Power amplifier
i. PCO Power change over
j. PJR Pulse jam rejection
k. PLL Phase locked loop
l. PSI Pod status indicator
m. RAT Ram air turbine
n. RF Radio frequency
o. RFI Radio frequency interference
p. SCP Signal-Control Processor
q. VCO Voltage controlled oscillator
r. VCTR Video cassette tape recorder

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6.6 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in DL 1064AS4100, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.7 Part or identifying number (PIN). The following is provided as supplemental information. Nomenclatures noted in Table XI were approved under ECP-1033 for AN/AWW-9B WRA's. In addition, Table XII provides cross-reference information showing the part numbers for the Signal Control-Processor wire.

TABLE XI. Nomenclatures approved under ECP-1033 for AN/AWW-9B WRAS.

Ref des	Nomenclature	Part number
UNIT 1	Guided Weapon Control-Monitor Set AN/AWW-9B	1064AS4100
1E1/ 1E2	(FWD/AFT) Antenna Assembly AS-3504A/ AWW-9 or AS-3504/AWW-9 (alternate)	1064AS1470 or 1064AS470 (alternate)
1A2	Signal Control-Processor C-1233/AWW-9B	1064AS4110
1A3	Radio Transmitter T1379A/AWW-9	1064AS2139
1A4	Video Cassette Tape Recorder RO-570/AXQ or RO-518/AXQ (alternate)	1342AS180 or V-1000AB-R
1A5	Transmitter Power Supply PP-7452A/AWW-9	1064AS3150
1A6	Filter-Circuit Breaker F-1514/AWW-9	1064AS300
1A7	Single Demodulator MD-1114/AWW-9	1064AS166
1RE1	Television Receiver R-2095A/AWW-9	1064AS2500 or 1064AS500-2 (alternate)

NOTE: Part of the function of ECP-1033 is to clean up previous configuration control problems with formally approved nomenclatures, which the table above reflects.

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TABLE XII. Signal Control Processor wire list.

Wire number	From	To	Wire part number
1	J1-	K1-D2	M22759/9-24-3
2	J1-2	K1-C2	M22759/9-24-5
3	J1-3	K1-B2	M22759/9-24-6
4	J1-4	E2-2	M22795/9-24-9
5	J1-9	E2-2	M22759/9-24-9
6	XA7-5	XA6-14	200AS100-22
7	J2-5	XA17-19	M22759/9-24-9
8	E2-2	M1-2	M22759/9-24-9
9	2-1	T1-BLK	SELF-LEAD
10	S5-1NC	ME7	M22759/9-24-92
11	J1-17	E1	M22759/9-24-0
12	J1-26	E1	M22759/9-24-0
13	J2-7	E1	M22759/9-24-0
14	J1-6	K1-D1	M22759/9-24-3
15	J1-13	XA1-3	M22759/9-24-9
16	E3-1	T1-ORN	SELF-LEAD
17	E3-1	K1-D1	M22759/9-24-3
18	XA4-3	XA8-5	200AS100-24
19	J1-7	K1-C1	M22759/9-24-5
20	XA1-1	XA1-21	200AS100-24
21	E4-1	T1-GRN	SELF-LEAD
22	E4-1	K1-C1	M22759/9-24-5
23	XA5-3	XA9-5	200AS100-24
24	J1-8	E5-2	M22759/9-24-6
25	XA7-24	XA6-6	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
26	E5-1	T1-BLU	SELF-LEAD
27	E5-1	K1-B1	M22759/9-24-6
28	XA1-2	XA7-43	200AS100-24
29	E45	Q4-E	200AS100-24
30	E7	K1-X1	200AS100-22
31	E8	K1-X2	200AS100-22
32	E8	E6	200AS100-22
33	CR1-C	E7-1	SELF-LEAD
34	CR1-A	E8-1	SELF-LEAD
35	M1-1	E3-2	M22759/9-24-3
36	XA1-3	XA15-28	M22759/9-24-9
37	S6-5	XA7-33	M22759/9-24-0
38	J1-22	XA7-26	M22759/9-24-1
39	J1-25	XA1-4	M22759/9-24-8
40	J1-27	XA17-12	M22759/9-24-4
41	J1-28	XA17-36	200AS139-22E10
42	J1-29	SHIELD OF #41 AT J1	M22759/9-24-0
43	XA10-19	XA17-35	200AS139-22E10
44	XA1-42	XA1-4	200AS100-24
45	J8-1	XA7-19	M22759/9-24-1
46	SHIELD OF #43 AT XA17	SHIELD OF #41 AT XA17	M22759/9-24-0
47	XA15-3	XA7-34	M22759/9-24-92
48	J1-30	E27	M22759/9-24-8
49	XDS2-2	XA10-3	M22759/9-24-8

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
50	E27	XA17-1	M22759/9-24-8
51	J1-33	XZA1-2	M22759/9-24-96
52	XA1-2	XA17-2	200AS100-
53	J1-35	XA7-31	200AS139-22E10
54	J1-34	SHIELD OF #53 AT J1	M22759/9-24-0
55	S6-4	XA14-10	M22759/9-24-92
56	S6-3	XA17-32	200AS139-22E10
57	XA17-11	XA10-21	200AS139-22E10
58	XA12-12	XA13-25	200AS100-24
59	SHIELD OF #57 AT XA17	SHIELD OF #56 AT XA17	M22759/9-24-0
60	J1-37	XA11-5	M17/95-RG180
61	J1-18	SHIELD OF #60 AT J1	M22759/9-24-0
62	J1-19	SHIELD OF #60 AT J1	M22759/9-24-0
63	J1-36	SHIELD OF #60 AT J1	M22759/9-24-0
64	J1-38	SHIELD OF #60 AT J1	M22759/9-24-0
65	XA11-6	SHIELD OF #60 AT XA11	M22759/9-24-0
66	J6	XA11-5	M17/95-RG180
67	XA11-6	SHIELD OF #66 AT XA11	M22759/9-24-0
68	J1-39	XA6-19	M22759/9-24-5
69	XA6-19	XA13-43	200AS100-24
70	XA6-19	XA5-3	M22759/9-24-5
71	XA5-3	XA4-3	200AS100-24
72	XA8-12	XA9-12	200AS100-24
73	J1-40	XA6-23	M22759/9-24-6
74	XA6-23	XA11-10	M22759/9-24-6

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
75	XA16-8	XA1-16	M22759/9-24-91
76	XA11-10	XA2-8	M22759/9-24-6
77	XA2-8	XA13-21	200AS100-24
78	XA2-8	XA17-27	200AS100-24
79	XA5-4	XA8-13	200AS100-24
80	XA4-9	XA8-19	200AS100-24
81	XA6-23	XA5-6	M22759/9-24-6
82	XA8-7	XA9-7	200AS100-24
83	XA16-19	XA1-17	M22759/9-24-93
84	XA6-6	XA15-2	M22759/9-24-2
85	XA14-5	XA16-42	200AS100-24
86	J1-41	CR2-C	M22759/9-24-2
87	CR2-C	XA6-6	200AS100-24
88	XA7-2	XA6-6	200AS100-24
89	XA5-2	XA9-19	200AS100-24
90	CR2-C	XA1-20	M22759/9-24-2
91	XA5-2	XA12-7	M22759/9-24-5
92	XA12-7	XA13-44	200AS100-24
93	XA10-2	XA10-44	200AS100-24
94	XA8-3	XA5-6	200AS100-24
95	XA9-3	XA5-6	200AS100-24
96	XA5-2	XA4-9	200AS100-24
97	XA8-21	XA4-4	200AS100-
98	J2-1	XA17-4	M17/95-RG180

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
99	J2-2	SHIELD OF #98 AT J2	M22759/9-24-0
100	XA11-1	XA17-5	M17/95-RG180
101	SHIELD OF #98 AT XA17	SHIELD OF #100 AT XA17	M22759/9-24-0
102	SHIELD OF #100 AT XA11	XA11-2	M22759/9-24-0
103	J2-3	XA6-14	M22759/9-24-4
104	XA-6	XF1-2	M22759/9-24-7
105	J2-12	S1-1C	M22759/9-24-3
106	E26	XA7-1	200AS100-24
107	E26	S1-1C	M22759/9-24-3
108	XA6-16	XA2-6	200AS100-
109	XA-6	XA11-12	M22759/9-24-3
110	XA6-16	XA13-22	200AS100-24
111	XA6-6	XA17-28	200AS100-24
112	XA9-21	XA5-5	200AS100-24
113	XA11-12	XA5-5	200AS100-24
114	XA9-18	XA5-8	200AS100-24
115	XA8-15	XA9-15	200AS100-24
116	XA8-10	XA9-10	200AS100-24
117	XA5-5	XA4-4	200AS100-24
118	XA10-30	E9	200AS100-22
119	J2-11	S1-NO	M22759/9-24-6
120	XA7-25	S1-NO	M22759/9-24-6
121	XA17-23	S1-NO	M22759/9-24-6
122	XA1-4	XA17-24	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
123	T1-RED	XA6-1	SELF-LEAD
124	T1-RED	XA6-2	SELF-LEAD
125	T1-BRN	XA6-11	SELF-LEAD
126	T1-BRN	XA6-22	SELF-LEAD
127	T1-WHT/BLU	CR4-C	SELF-LEAD
128	CR4-C	CR7-A	200AS100-22
129	T1-WHT/ORN	CR5-C	SELF-LEAD
130	CR5-C	CR8-A	200AS100-22
131	T1-WHT/GEN	CR6-C	SELF-LEAD
132	CR6-C	CR9-A	200AS100-22
133	VR4-A	E11	200AS100-22
134	CR4-A	CR5-A	200AS100-22
135	CR5-A	CR6-A	200AS100-22
136	CR7-A	CR8-C	200AS100-22
137	CR8-A	CR9-C	200AS100-22
138	CR9-A	E14	200AS100-22
139	E14	E16	200AS100-22
140	E16	XF1-1	M22759/9-24-4
141	C1-POS	E14	SELF-LEAD
142	C1-NEG	E15	SELF-LEAD
143	R2	E16	SELF-LEAD
144	R2	E17	SELF-LEAD
145	E17	E15	200AS100-22
146	E15	E13	200AS100-22

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
147	XA1-10	XA7-3	200AS100-24
148	XA10-16	J8-20	M22759/9-24-92
149	XA1-10	XA10-7	M22759/9-24-91
150	XA11-4	XA7-6	M22759/9-24-95
151	XA10-10	XA11-4	200AS100-24
152	XA8-9	XA9-9	200AS100-24
153	XA10-8	E27	M22759/9-24-8
154	XA10-14	XA14-28	200AS100-24
155	R4	XA11-3	SELF-LEAD
156	R4	XA11-7	SELF-LEAD
157	XA11-7	CR11-C	SELF-LEAD
158	XA11-4	CR11-A	SELF-LEAD
159	XA11-7	XA16-43	200AS100-24
160	J8-2	XA7-18	M22759/9-24-93
161	J8-3	XA7-17	M22759/9-24-95
162	J8-4	XA7-16	M22759/9-24-96
163	XA10-12	XA1-6	M22759/9-24-9
164	XA1-6	XA14-6	200AS100-24
165	XA10-11	XA1-8	M22759/9-24-93
166	XA1-8	XA14-25	200AS100-24
167	XA7-44	E11	200AS100-22
168	XA1-6	XA7-8	200AS100-24
169	XA6-9	Q2-E	200AS100-22
170	XA6-10	Q2-C	200AS100-22

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
171	XA6-7	Q2-B	200AS100-22
172	XA6-5	Q1-B	M22759/9-24-91
173	XA6-4	Q1-E	M22759/9-24-92
174	XA6-3	Q1-C	M22759/9-24-95
175	XA6-13	Q3-B	M22759/9-24-91
176	XA6-18	Q3-C	M22759/9-24-92
177	XA6-20	Q3-E	M22759/9-24-95
178	CR13-C	E26	M22759/9-24-3
179	XA15-1	XA16-23	200AS100-24
180	XA14-1	XA15-23	200AS100-24
181	XA6-15	CR3-A	200AS100-24
182	XA6-17	R1-1	M22759/9-24-96R
183	XA6-21	R1-2	M22759/9-24-96
184	XA6-12	E11	200AS100-22
185	J1-31	XA17-30	M22759/9-24-4
186	J3	XA5-1	M17/95-RG180
187	SHIELD OF #186 AT XA5	XA5-7	M22759/9-24-0
188	XA5-4	XA4-1	200AS100-24
189	XA8-1	E13	200AS100-22
190	XA9-1	E-13	200AS100-22
191	XA5-1	XA4-7	200AS100-22
192	XA10-1	XA11-10	200AS100-24
193	XA4-7	E21	200AS100-22
194	XA11-11	E12	200AS100-22

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
195	XA10-22	XA11-8	200AS100-24
196	XA10-13	XA11-3	200AS100-24
197	XA10-4	E26	M22759/9-24-3
198	XA5-9	XA8-18	200AS100-24
199	XA1-18	XA15-17	M22759/9-24-92
200	XA5-9	XA5-8	200AS100-24
201	XA5-10	XA4-5	200AS100-24
202	XA5-11	XA4-2	200AS100-24
203	J7	XA5-12	M17/95-RG180
204	SHIELD OF #302 AT XA5	E21	M22759/9-24-0
205	XA1-10	XA14-2	200AS100-24
206	XA1-19	J8-25	M22759/9-24-91
207	XA1-13	XA16-31	M22759/9-24-93
208	XA1-12	XA1-13	200AS100-24
209	XA2-12	XA2-23	200AS100-22
210	XA6-8	XA5-7	M22759/9-24-0
211	E44	XA2-12	200AS100-22
212	XA2-23	XA12-22	200AS100-22
213	XA1-18	XA14-41	200AS100-24
214	XA14-34	XA10-20	200AS100-24
215	J8-23	XA1-13	M22759/9-24-93
216	XA1-8	XA7-7	200AS100-24
217	SDX1-2	E1	200AS100-22
218	SDX1-1	XF1-2	M22759/9-24-4

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
219	SDX1-1	XDS2-1	200AS100-22
220	R3	E8	SELF-LEAD
221	R3	E18	SELF-LEAD
222	J1-12	E18	M22759/9-24-96
223	J2-8	E18	M22759/9-24-96
224	TB1-E2	XA13-41	M22759/9-24-92
225	TB1-E3	XA13-40	M22759/9-24-93
226	TB1-E4	XA13-18	M22759/9-24-91
227	TB1-E5	XA13-19	M22759/9-24-95
228	TB1-E6	XA12-21	M22759/9-24-96
229	TB1-E7	XA12-13	M22759/9-24-8
230	TB1-E8	XA12-23	M22759/9-24-9
231	TB1-E9	XA13-17	M22759/9-24-7
232	TB1-E10	XA13-9	M22759/9-24-91
233	XA13-2	XA8-16	M22759/9-24-9
234	XA9-13	XA12-4	M22759/9-24-92
235	XA13-1	XA10-6	M22759/9-24-95
236	XA12-6	XA13-39	200AS100-24
237	XA12-8	XA13-7	200AS100-24
238	XA12-9	XA13-38	200AS100-24
239	XA12-10	XA13-29	200AS100-24
240	XA12-11	XA13-24	200AS100-24
241	XA12-12	XA13-23	200AS100-24
242	XA12-18	XA13-8	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
243	XA2-5	XA13-30	200AS100-24
244	XA17-41	XA13-28	200AS100-24
245	XA17-20	XA13-27	200AS100-24
246	XA17-3	XA13-26	200AS100-24
247	E46	XA13-15	M22759/9-24-7
248	XA17-25	XA13-10	200AS100-24
249	XA17-19	XA13-6	200AS100-24
250	XA17-42	XA13-5	200AS100-24
251	XA17-9	XA1-1	200AS100-24
252	J1-24	XA1-1	M22759/9-24-7
253	J1-21	XA7-9	M22759/9-24-9
254	J1-20	XA7-20	M22759/9-24-91
255	XA12-1	XA17-13	200AS100-24
256	XA12-4	XA17-21	200AS100-24
257	XA12-5	XA17-18	200AS100-24
258	XA12-17	XA17-40	200AS100-24
259	XA12-16	XA17-15	200AS100-24
260	S2-2	XA12-16	M22759/9-24-2
261	S2-2	S-2	200AS100-24
262	S3-2	S-2	200AS100-24
263	XA2-2	XA17-33	200AS100-24
264	XA2-22	XA17-14	200AS100-24
265	XA2-14	XA17-39	200AS100-24
266	XA2-16	XA17-37	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
267	XA2-17	XA17-17	200AS100-24
268	XA2-18	XA17-16	200AS100-24
269	XA2-20	XA17-38	200AS100-24
270	XA11-9	XA17-34	200AS100-24
272	XA1-19	XA14-36	200AS100-24
273	XA16-28	XA1-12	M22759/9-24-95
274	XA16-6	XAE1-15	M22759/9-24-8
275	XA14-44	E19	200AS100-22
276	XA14-22	E19	200AS100-22
277	XA15-44	E20	200AS100-22
278	XA15-22	E20	200AS100-22
279	XA16-44	E20	200AS100-22
280	XA16-22	E20	200AS100-22
281	XA14-9	XA16-10	200AS100-24
282	XA14-27	XA16-37	200AS100-24
283	XA14-17	XA16-41	200AS100-24
284	XA14-20	XA16-25	200AS100-24
285	XA14-19	XA16-27	200AS100-24
286	XA1-18	XA16-32	M22759/9-24-92
287	CR15-A	E47	SELF-LEAD
288	SHIELD OF #56 AT S6	E6	M22759/9-24-0
289	XA14-13	XA1-11	200AS100-24
290	XA15-19	XA1-12	M22759/9-24-93
291	XA14-39	XA1-12	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
292	XA15-36	XA17-8	M22759/9-24-95
293	E47	XA12-14	M22759/9-24-4
294	S3-1	XA12-3	M22759/9-24-9
295	S4-1	XA12-20	M22759/9-24-96
296	S2-1	E19	M22759/9-24-8
297	XA17-44	XA2-4	200AS100-22
298	J4	XA17-26	M17/95-RG180
299	XA2-4	XA2-12	M17-95-RG180
300	SHIELD OF #299 AT XA2		M22759/9-24-0
301	J5	XA17-6	M17/95-RG180
302	XA12-2	XA13-20	200AS100-24
303	XA2-10	XA17-10	200AS100-24
304	XA2-10	XA12-2	200AS100-24
305	XA12-19	XA13-3	200AS100-24
306	XA12-19	XA2-10	200AS100-24
307	XA16-7	XA1-14	M22759/9-24-96
308	TB1-E1	XA2-10	M22759/9-24-1
309	XA14-15	XA1-14	200AS100-24
310	XA15-33	XA1-14	M22759/9-24-96
311	XA14-34	XA1-15	200AS100-24
312	XA15-11	XA1-15	M22759/9-248-
313	XA14-40	XA1-16	200AS100-24
314	XA15-43	XA1-16	M22759/9-24-91
315	XA14-16	XA1-17	200AS100-24

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
316	XA15-21	XA1-17	M22759/9-24-93
317	CR15-C	E45	SELF-LEAD
318	XA14-35	XA16-29	200AS100-24
319	XA13-42	E19	200AS100-22
320	XA14-11	XA16-9	200AS100-24
321	XA14-26	XA15-25	200AS100-24
322	XA14-14	XA15-4	200AS100-24
323	J8-5	XA7-15	M22759/9-24-92
324	J8-6	XA7-14	M22759/9-24-93
325	J8-7	XA7-13	M22759/9-24-95
326	J8-8	XA7-12	M22759/9-24-8
327	J8-9	XA7-11	M22759/9-24-9
328	J8-10	XA7-10	M22759/9-24-91
329	J8-11	E1	M22759/9-24-0
330	XA14-24	XA15-26	200AS100-24
331	XA14-18	XA15-15	200AS100-24
332	XA1-19	XA15-27	M22759/9-24-91
333	XA14-41	XA15-5	200AS100-24
334	E26	XA14-23	M22759/9-24-3
335	S6-1	XA14-33	200AS139-22E10
336	SHIELD OF #335 AT S6	E6	M22759/9-24-0
337	CR13-C	XA6-16	M22759/9-24-3
338	E46	XA10-17	M22759/9-24-7
339	XA10-27	E47	M22759/9-24-4
340	XA10-15	Q2-B	200AS100-24
341	XA14-12	XA1-3	200AS100-24
342	Q4-C	S5-1C	M22759/9-24-96

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
343	B1-J3	E3-2	M22759/9-24-3
344	B1-J3	E4-2	M22759/9-24-5
345	B1-J4	E5-2	M22759/9-24-6
346	CR14-C	E45	SELF-LEAD
347	E16	E22	200AS100-22
348	E22	E24	M22759/9-24-9
349	E23	E13	200AS100-22
350	E25	E13	M22759/9-24-9
351	C2 POS	E22	SELF-LEAD
352	C2 NEG	E23	SELF-LEAD
353	C3 POS	E24	SELF-LEAD
354	C3 NEG	E25	SELF-LEAD
355	XA15-7	J8-24	M22759/9-24-8
356	XA15-41	XA16-39	200AS100-24
357	XA15-42	XA16-38	200AS100-24
358	XA15-20	XA16-40	200AS100-24
359	J1-16	E1	M22759/9-24-0
360	J1-10	E1	M22759/9-24-0
361	E27	XA7-27	200AS100-24
362	J8-24	XA16-36	M22759/9-24-8
363	E46	XDS2-1	M22759/9-24-2
364	XA1-20	XA11-3	M22759/9-24-9
365	XA1-20	XA12-7	200AS100-24
366	XA1-20	XA14-7	200AS100-24
367	J1-11	XA7-35	M22759/9-24-91
368	SHIELD OF #298 AT XA2	XA2-23	M22759/9-24-0
369	S6-4	XA15-30	M22750/9-24-0

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
370	CR14-A	E46	SELF-LEAD
371	XA15-24	XA7-32	200AS139-22E10
372	SHIELD OF #371 AT XA7	XA7-22	M22759/9-24-0
373	S6-2	XA7-3	M22759/9-24-92
374	XA15-10	J8-12	M22759/9-24-93
375	XA15-31	J8-13	M22759/9-24-95
376	XA15-9	J8-14	M22759/9-24-96
377	XA15-32	J8-15	M22759/9-24-91
378	XA15-8	J8-16	M22759/9-24-93
379	XA15-12	J8-17	M22759/9-24-95
381	XA16-24	J8-1	M22759/9-24-96
382	XA16-3	J8-2	M22759/9-24-8
383	XA7-4	J8-19	M22759/9-24-91
384	J8-18	XA3-8	200AS139-22E10
385	J1-23	E47	M22759/9-24-4
387	J8-3	XA16-30	M22759/9-24-95
388	J8-4	XA16-2	M22759/9-24-96
389	J8-5	XA16-16	M22759/9-24-92
390	J8-6	XA16-17	M22759/9-24-93
391	J8-7	XA16-18	M22759/9-24-95
392	J8-8	XA16-15	M22759/9-24-8
393	J8-9	XA16-21	M22759/9-24-9
394	J8-10	XA16-20	M22759/9-24-91
398	SHIELD OF #384 AT J8	J8-21	M22759/9-24-0
399	XA10-26	XA3-12	200AS100-24
400	XA10-23	XA3-10	200AS100-24
401	XA3-11	E9	200AS100-22

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TABLE XII. Signal Control Processor wire list - Continued.

Wire number	From	To	Wire part number
402	XA10-41	XA3-9	200AS139-22E10
404	SHIELD OF WIRE #402 at XA3	XA3-11	M22759/9-24-0
405	XA10-43	XA3-8	200AS100-24
406	XA10-25	XA3-7	200AS139-22E10
408	SHIELD OF WIRE #406 AT XA3	XA3-11	M22759/9-24-0
409	XA1-5	J1-15	----------
410	XA1-7	E44	----------

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6.8 Subject term (keyword) listing.

Control-Monitor
Data link
Decoding
DPSK
FSK
Guided weapon
Modulator, half-sine
Pod
Processor, signal-control
Radome
Receiver, television
Recorder, video cassette
Weapon system
WALLEYE
SLAM

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - AS
(Project 1430-N148)

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3. DOCUMENT TITLE
CONTROL-MONITOR SET, GUIDED WEAPON, AN/AWW-9B, P/N 1064AS4100

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST, SYSTEMS REQUIREMENTS DEPARTMENT	(1) Commercial (908) 323-7488	(2) AUTOVON 624-7488
c. ADDRESS (Include Zip Code) CODE SR3, HIGHWAY 547, LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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